

Telecommunications Engineer Cycle
Option : MIT

End of Studies Project Report

Implementation and development of an MES monitoring and alerting application in an industry 4.0

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Tunis, the XXX

Naoufel Ben Refifa

Abstract

Industry 4.0 has been gaining in popularity and use in recent years as it's considered as one of the most significant transformation in the way products are produced especially when highlighting the digitization that the manufacturing field is witnessing. In fact, this transformation is so compelling that it is being called Industry 4.0 to represent the fourth revolution that has occurred in manufacturing. From the first industrial revolution, mechanization through water and steam power, and mass production and assembly lines using electricity in the second, to early digitizing through semiconductors, mainframe computing, personal computing and the Internet in the third, Industry 4.0 is taking what was started in the previous revolution and enhancing it with smart and autonomous systems fueled by data and machine learning. With that being said, this field depends heavily on data, so to be able to handle this transformation one should be able to have control over the data and be creative when dealing with it.

One of the leading companies to follow this transformation is SAP (Systems Applications and Products) through its manufacturing execution system (MES) which provides real-time data and analytics of your manufacturing floor. This software executes manufacturing processes, analyzes manufacturing and business data, and integrates systems based on Industry 4.0. SAP Digital Manufacturing is designed to track and enhance manufacturing performance, improve visibility and reporting consistency, gain faster and more consistent insights, adapt to individualization requirements, and put the business assets to work more effectively.

In this report, an Introduction chapter (Chapter.1) is dedicated for the reader to present the hosting company and to get familiarised with the necessary terms to understand the problematic that we started from and how we decided to try solving it. Then, in Chapter.2, a more detailed descriptions of the technical fields and the domain of application and also, the reader can find the most useful definitions of terms that will be used later in this report. The following chapter, Chapter.3, is dedicated for the application design and development needed to be realised during this project to give a better idea about this project. Afterwards, we stated the results and implementations of our work in Chapter.4, which contains also some descriptions about this achieved results.

And finally, a general conclusion (Chapter.5) to sum up the work done through out this internship and state the possible opportunities to make the results of this project even better.

Key words : SAP, SAP MII, SAP PCo, Kepware, SAPUI5, industry 4.0

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Chapter 1

Introduction

In this chapter, I will be presenting Wynsys, the company in which I have conducted my internship. Also, i think it's necessary before going deep into details to set the problematic and give a general view about the solution proposed and what our project will be in few words.

1.1 Wynsys: Hosting Company Introduction

1.1.1 General introduction

Created in 2007, WYNSYS has established itself as one of the most efficient SAP integrator. Thanks to its experience, expertise in SAP products and a highly qualified team based in Tunis and Paris, it is well positioned to provide high added value SAP projects and efficient maintenance and support services whatever the location of its client. Adopting a perspective of continuous evolution, it has become in 2012, a SAP partner re-seller obtained the highly coveted and recognized certification "SAP Partner Center Of Expertise".

In January 2013, WYNSYS was rated by international bureaus as "SAP Gold Partner "and in 2015 became a member of the SAP United VARs network, the largest alliance SAP partners. Indeed, this development leads to an average growth in turnover of 20 per cent on past five years.

Today, its workforce exceeds 400 employees, including 250 consultants technical and functional.[**1**]

1.1.2 WYNSYS clients

WYNSYS is considered a benchmark player in the field of consulting and integration ERP solutions, it has succeeded in winning the trust of various national companies and international thanks to the guaranteed services it offers. Among the customers of WYNSYS, we finds Japanese automotive supplier YAZAKI, French multinational Sanofi at scale international, the Tunisian company Sancella Tunisie, the agro-food company Landor at national scale.



Figure 1.1: WYNSYS clients

1.1.3 Field of activity and company services

WYNSYS is a company that operates in the field of IT consulting and engineering in particular on various integration projects, deployment and support of business solutions. By relying on the know-how, experience and commitment of the entire team in the mastery of deadlines, budgets and the quality of the services offered to its clients, the selective manner of recruitment engineers from major Tunisian and French schools, WYNSYS is now presented as a benchmark player in the field of computer engineering. Indeed, WYNSYS offers three types of services: —Consulting and Integration: from a technical and functional skills center specializing in the integration, deployment, start-up assistance and maintenance of business solutions. —Application maintenance: the WYNSYS teams ensure the proper functioning of applications in a process of industrialization and control of maintenance budgets thanks to the services of TMA, —Test Services: By providing testing services designed to help customers reduce IT costs and enhance the performance of ERP applications and their reliability.

1.1.4 Organization of the WYNSYS Company

The operational organization of WYNSYS is devided into a matrix structure based on functional and divisional units: —The functional units are the WYNSYS personnel who are distributed according to a criterion of specializing in SAP modules, they are at the service of divisional units. —Divisional units are company projects that call for the assistance of functional units.[**1**]

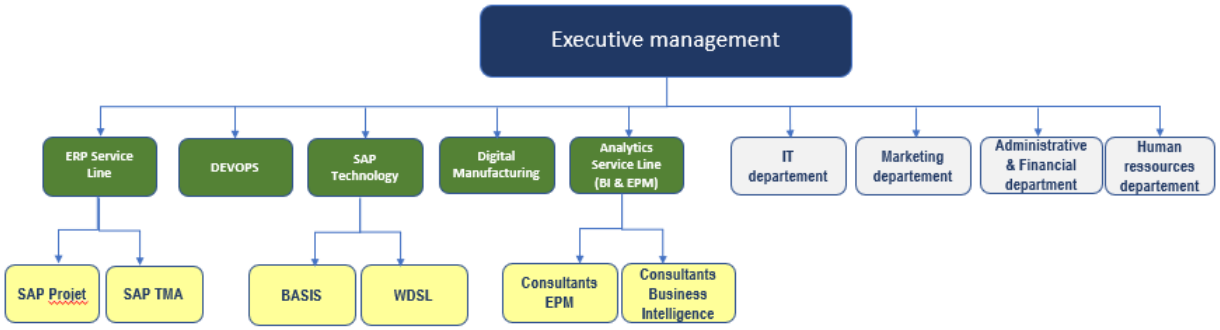


Figure 1.2: WYNSYS Organizational Chart

During this project, the digital manufacturing department will be the one concerned by the results of this work.

1.2 Problematic

To set the problematic in a clear way, let's start from a real daily situation that any of the industrial companies can face. A shop floor worker is trying to reload a certain machine with a predefined quantity of materials which is considered as a daily task for this worker, but for some reason the production isn't taking its way and the chain of work is in cut. this worker doesn't have the requirements to define the real problem and only has to report back to his superior and this reporting and examination steps afterwards can take even a day or more which s surely a big loss to the company. One of the major problems that can be the cause of this cut of work is the difficulty to be updated with every alert either on the infrastructure level or on the digital level which means the applications that monitor the working machines and in a general way the shop floor.

1.3 Solution

When having a look at the given problematic, maybe the best solution to propose is to settle a monitoring and alerting system within any industrial company to ensure that any problem is reported in no time and and everything is under control and so the actions can be taken immediately and as a result the work

will be moving forward smoothly without interruptions and even if there's a cut it will take a lot less time than the situation where the problematic is set. to make things easier and more accessible for either the higher ups within the industry or even the simple worker, a friendly user application will be the best call for this problematic and so it will be easily accessed and a clear overview over the anomalies and the status of the shop floor infrastructure is shown.

1.4 Project Overview

During this project, we will try to implement and develop an application which will has as a role the monitoring and alerting within a manufacturing execution system which provides information that make easier to manufacturing decision makers understanding the current conditions on the plant floor. And so this application will be a big help to the industrial workers to smoothly have access to the details of any given problem and as a result define exactly the steps that have to be taken to get back to work int fastest possible way.

All along this chapter, i have tried to give a simple overview about the idea of this project, what was the initial problem and what solution did we estimated as efficient. The following chapter, is a deeper dive in both the technical field and business aspect of the work in which I will be presenting the terminology and cite the necessary vocabulary in order to ensure a more unclouded reading.

Chapter 2

Background

This chapter will present us most of the terms and define them so that the reader will find no problem of understanding when going deep in the technical parts of this project.

2.1 Industry 4.0

2.1.1 Definition

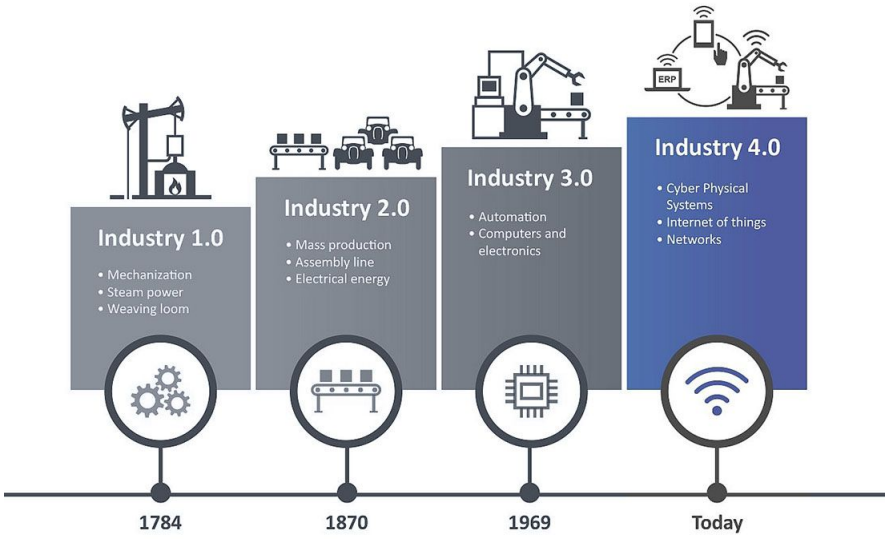


Figure 2.1: Industry Revolutions

Industry 4.0 can be presented as the combination of digital technology breakthroughs, including robots and artificial intelligence, advanced sensors, cloud computing, the Internet of Things (IoT), smartphone compatibility, and a variety of other uses.[**2**].

Manufacturing procedures and techniques have evolved through decades, assisting firms in improving their production, performance, and output[**2**]. Following the lean revolution of the 1970s, outsourcing in the 1990s, and automation in the 2000s, Industry 4.0 is the fourth contemporary industrial upheaval. With that being said, this revolution is pushing humanity forward with more advanced techniques in manufacturing[**2**].

2.1.2 Principles

As any previous revolution, industry 4.0 has some principles that we have to mention

1.Interconnection:

You may now link machines, sensors, and other equipment to the personnel entrusted with monitoring your process for efficacy and efficiency using wireless communication technology and the capabilities of the Internet of Things.[**2**]

2.Information Transparency:

The openness provided by Industry 4.0 technology offers operators with a big load of data to help them make informed decisions. Interconnectivity allows operators to acquire massive volumes of data and information from all points in the production process, allowing them to increase functionality and discover important areas where innovation and development are needed.[**2**]

3.Decentralized Decisions:

Operators may make choices both within and outside of production facilities because to interconnection and information openness. This capacity to bring together local and global data at the same time aids in better decision-making and overall productivity.[**2**]

4.Technical Assistance:

Humans' role in Industry 4.0 moves from machine operator to problem solver and decision maker. Assistance systems are intended to assist operators who must make quick judgments in order to handle critical situations.[**2**]

2.1.3 Contributions

Industry 4.0 can impact the process in which you are included in many ways :

The implementation of Industry 4.0 ideas may definitely provide firms with improved efficiency in their operations by connecting automation with data gathering and exchange methods.

Streamlining procedures and providing access to useful data can assist increase production while reducing resource use. Industry 4.0 also aids enterprises in increasing efficiency and revenue growth by saving money on materials and labor, as well as reducing customer rejection and production failures.

Industry 4.0 can also have an influence on production by facilitating tighter consumer relations. Technology, data, and information that may assist alter industrial operations may also improve the responsiveness of processes and systems to consumer demands. Interconnected technologies' unique characteristics enable manufacturers to respond and adapt to client requirements more rapidly, and even generate custom orders with less labor and setup time than traditional production.

Manufacturing processes are always changing, resulting in leaner production and more efficient procedures. You can guarantee that your processes are giving the essential information to guide improvements and optimal performance, keeping you ahead of the competition, with Industry 4.0 technology and strategy.

2.2 SAP : Shop floor

Manufacturing is more advanced than it has ever been. There are too many moving pieces between staff blunders, machine breakdowns, late shipment, and rush orders. It's impossible to avoid making mistakes if you don't plan ahead.

With the changing manufacturing scene, wise manufacturers are fast embracing the most up-to-date business process optimization methods, with shop floor management being arguably the most essential on the list.[**3**]



Figure 2.2: Why Shop Floor Management

2.2.1 Definition

A shop floor is where assembly or production takes place. This might be done manually by employees or with the help of modern automated technologies.

The term "shop floor management" refers to management at the place where things happen. Managers and production workers gather every day at this point of value creation to thoroughly examine the equipment, inventory, storage, and manufacturing processes in order to detect any disruptions or deviations in the value-added processes — and, if required, to take countermeasures.

The shop floor management meetings, also known as daily meetings, are an important element of the management process.[**3**]

SAP on the Shop Floor

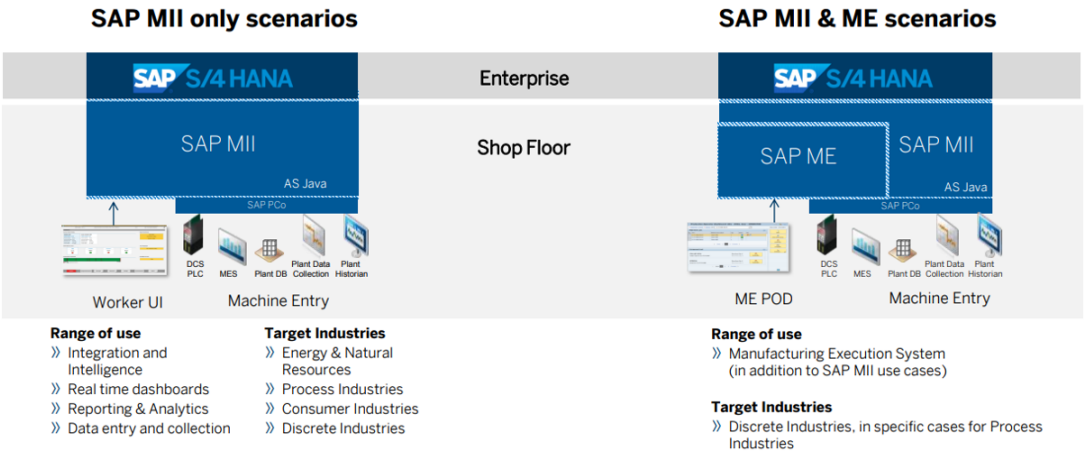


Figure 2.3: SAP on the Shop Floor Today

2.2.2 Objectives

An ideal shop floor management plan must take into consideration the following objectives:

- Ensure that supplies are routed properly on the shop floor.
- Effectively channelize shop floor processes and procedures
- Schedule materials, workers, other resources, and operations
- Identify and close communication gaps
- Keep an eye out for any deviations from established procedures and processes, and act quickly to remedy them.[**3**]

2.2.3 Benefits

Organizations that effectively manage their shop floors begin to see significant results in their production and assembly lines. That's because having a solid strategy in place makes it much easier to prevent issues and deal with them effectively.[**3**]

To truly understand the impact of shop floor management, organizations need to consider the following benefits:

Fewer Errors

The appropriate shop floor management enables for a smooth and quick flow of information, with IIoT solutions allowing for real-time connectivity.[**3**]

The production manager and other key decision-makers can keep up to date on ground operations and have total control over the supply chain thanks to the quick and transparent communication. [**3**]

As a consequence, recognizing and improving any problematic areas becomes second nature, resulting in a considerable reduction in mistakes and more reliable execution of regular procedures and processes.[**3**]

Reduction of Waste and Overhead Costs

As production cycles become more efficient, companies see a reduction in waste and overhead – and, as a result, an increase in profitability. [**3**]

This is made feasible by a reduction in mistakes and an overall increase in productivity at each stage of the manufacturing process.[**3**]

Time Savings

Project managers may make data-backed choices in rapid time thanks to automated access to crucial supply chain data in real-time via a floor management dashboard. [**3**]

As a consequence, project managers may save time and concentrate on other important tasks. On top of the improved shop floor management, this inevitably leads to a boost in other areas. [**3**]

2.3 SAP Manufacturing Execution System

2.3.1 From Papers to MES

Manufacturing is becoming more competitive and complicated, with customers demanding shorter lead times, cheaper production costs, and strict adherence to industry norms and regulations. Although it is critical to use real-time manufacturing data to drive business decisions, 85 percent of manufacturers still manage their production floor with paper and spreadsheets.[**5**]

ERP and other business systems collect order input and invoice output; nevertheless, the insights these systems give in terms of production are too shallow and inaccurate to improve operations. [**5**]

2.3.2 What is MES

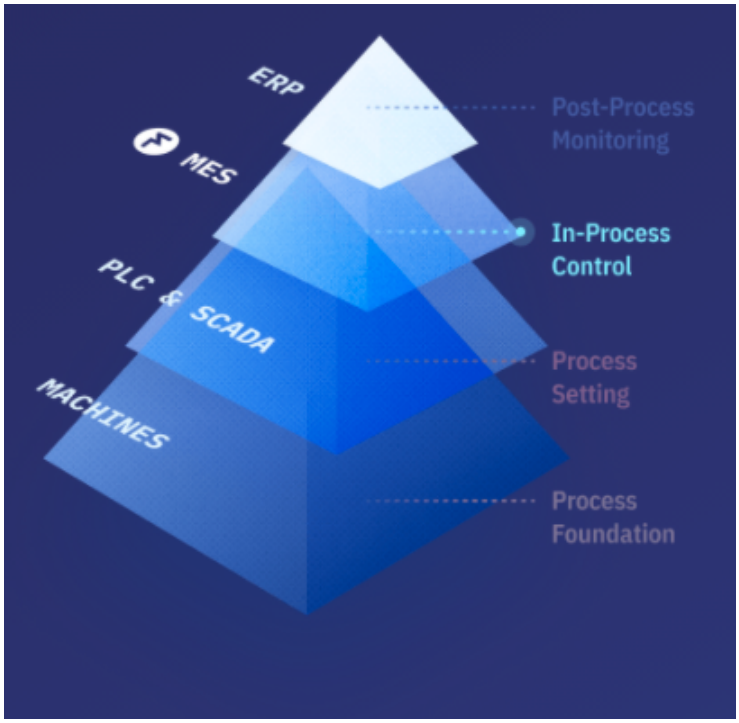


Figure 2.4: The Manufacturing Execution System

A Manufacturing Execution System, or MES, is a productivity solution that helps you organize and track all of your production orders, allowing you to deliver on time and optimize processes.[**5**]

It can link to your existing systems to provide you a real-time picture of your operations, making communication between managers easier, and it can report on KPIs so you can monitor operator and business performance.[**5**]

SAP MES Manufacturing Suite

SAP ME (SAP Manufacturing Execution) An entirety configurable MES focusing on discrete manufacturing processes and also a comprehensive standard integration with SAP ERP.

SAP MII (SAP Manufacturing Integration and Intelligence) can either be a stand-alone platform for industrial manufacturing operations man-

agement or be used for integration, functional and UI extension in bundle with SAP ME.

SAP Plant Connectivity (Pco) Connect machines to the MES, enterprise systems and the Industrial Internet of Things (IoT).

Digital Manufacturing Insight (DMI) Complements and extends SAP's existing on premise solution portfolio for digital manufacturing and also a cloud-based offering provides real-time, end-to-end visibility and manufacturing operations

To make it easier, let's have a look at the SAP digital manufacturing suite to comprehend how this different parts are reacting together.

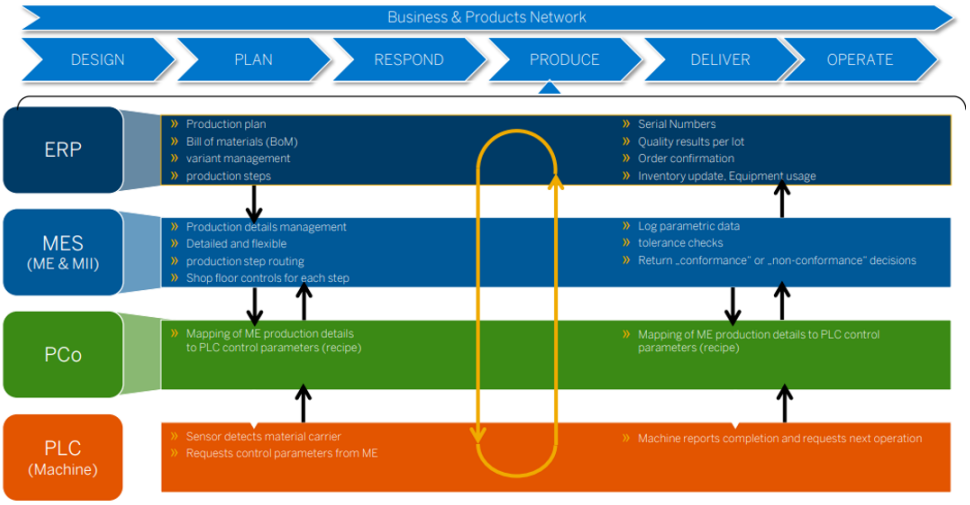


Figure 2.5: SAP Digital Manufacturing Architecture

2.3.3 Benefits of MES

Benefits to Operations

Gain Real-Time Visibility into Production Replace paper and spreadsheets with a visual dashboard showing you how production is proceeding in real time.[**5**]

Analyze Cycle Times and Improve Predictability Calculate cycle times to enhance forecasting and offer clients with precise available-to-promise dates.[**5**]

Identify Errors and Bottlenecks Immediately Be notified as soon as an error occurs, and rapidly discover and fix the root cause. Analyze production to identify bottlenecks and improve efficiency.[**5**]

Measure Labor Utilisation and Cost Measure the number of man-hours and costs involved in creating a product or providing customer service.[**5**]

Benefits to Finance

Enable Activity-Based Costing With a precise picture of overall costs, including granular labor usage, make critical pricing, production, and investment choices.[**5**]

Understand Marginal Cost Understand every component that goes into your production process to accurately price products and services.[**5**]

Maximize Impact of Cash-to-Cash Cycle Understanding your cycle durations and throughput will help you reduce the amount of work in process.[**5**]

Reduce Non-Value Adding Time Determine the core cause of production stopping problems in real time and identify bottlenecks to improve operations.[**5**]

Measure Labor Utilisation Understand your labor costs in order to get a realistic cost per unit generated.[**5**]

Foster Accountability Measure the performance of your team as well as the outcomes of important projects to make data-driven choices and keep your team responsible.[**5**]

Benefits to IT

Robust API The MES API is designed to never be the limiting factor in integrations, allowing you to link multiple systems and databases to the MES application.[**5**]

Security Data in MES is encrypted from beginning to end, both at rest and in transit.

Hardware-Agnostic Through a web browser or our API, MES may be accessed by any device, providing connection with devices ranging from consumer to enterprise-grade.[**5**]

Future-Proof When upgrading your ERP or implementing business systems, MES will never be an obstacle to integrating with them.[**5**]

2.3.4 Where is MES Used ?

Traceability "Active Quality Assurance for Zero-Defect Production"

- All work step are precisely controlled
- Batch and Serial Number tracking
- Historical build record

Time Sensitive Materials

- SAP Control of Engineering Master data
- Material Control and Usage
- Automatically apply holds and releases
- Check In and Out of materials
- Operator Notification

SPC (Statistical Process Control) with MII iSPC Charts

- SAP Perform real-time SPC analysis
- Check for trends and variance
- View the SPC analytical
- Trend alarming
- Multiple SPC chart types
- Improved quality through analysis
- Improve product and process through analytics

Routing Engine and Work Instruction

- Controls the flow of material and product through the manufacturing process
- Apply logic and controls based on real-time data
- Automatically route products based on business rules and results
- Integrated into ECC (ERP central component) for achievement reporting and updated routings
- Enforce the manufacturing process based on real-time condition monitoring
- Model multiple scenarios in a single routing, reducing engineering effort

Mobility

- Deliver browser-based mobile apps for wireless bar code scanners used on the shop floor to carry out manufacturing tasks.
- Allow production operations in the application to be carried out via mobile data terminals.
- Deliver as an exposed templates to aid in the building of customised mobile apps.

Management and Role Controls

- Assure task, access and user permission controls
- Manage skill certification centrally
- Extend certification temporarily
- Alerts on expiry or attempts to by-pass

2.4 SAP Manufacturing Integration and Intelligence

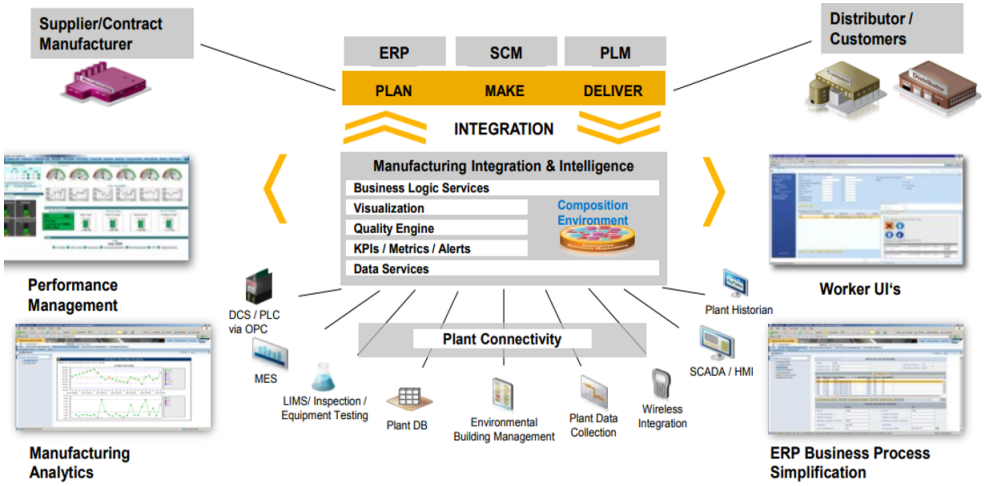


Figure 2.6: The Manufacturing Integration and Intelligence

2.4.1 What is SAP MII

SAP Manufacturing Integration and Intelligence (SAP MII) is a well-known platform for Manufacturing Execution (MES) and integration solutions that has been proved in the process sector. SAP MII enables real-time production monitoring and offers powerful data analysis tools.

Data from a variety of systems may be linked and merged with data from other business systems using SAP MII. This connection gives comprehensive insights into measures from SAP ERP all the way down to the shop floor, as well as across shop floor processes and apps. These insights assist in making decisions about equipment utilization and adjusting production priorities and also ensures necessary actions can be executed automatically or manually.

SAP MII solutions take on the role of core component and orchestrator for data-driven, evidence-based decision making. MII collects and processes data from all important business systems before presenting it to key stakeholders and decision makers as integrated data. SAP MII, for example, may be used to monitor, gather, measure, and compare the performance of plants and assets using various automation system infrastructures. SAP Plant Connectivity (SAP PCo) can be used to link factory floor equipment for data gathering and assessment to allow more relevant information.[**4**]

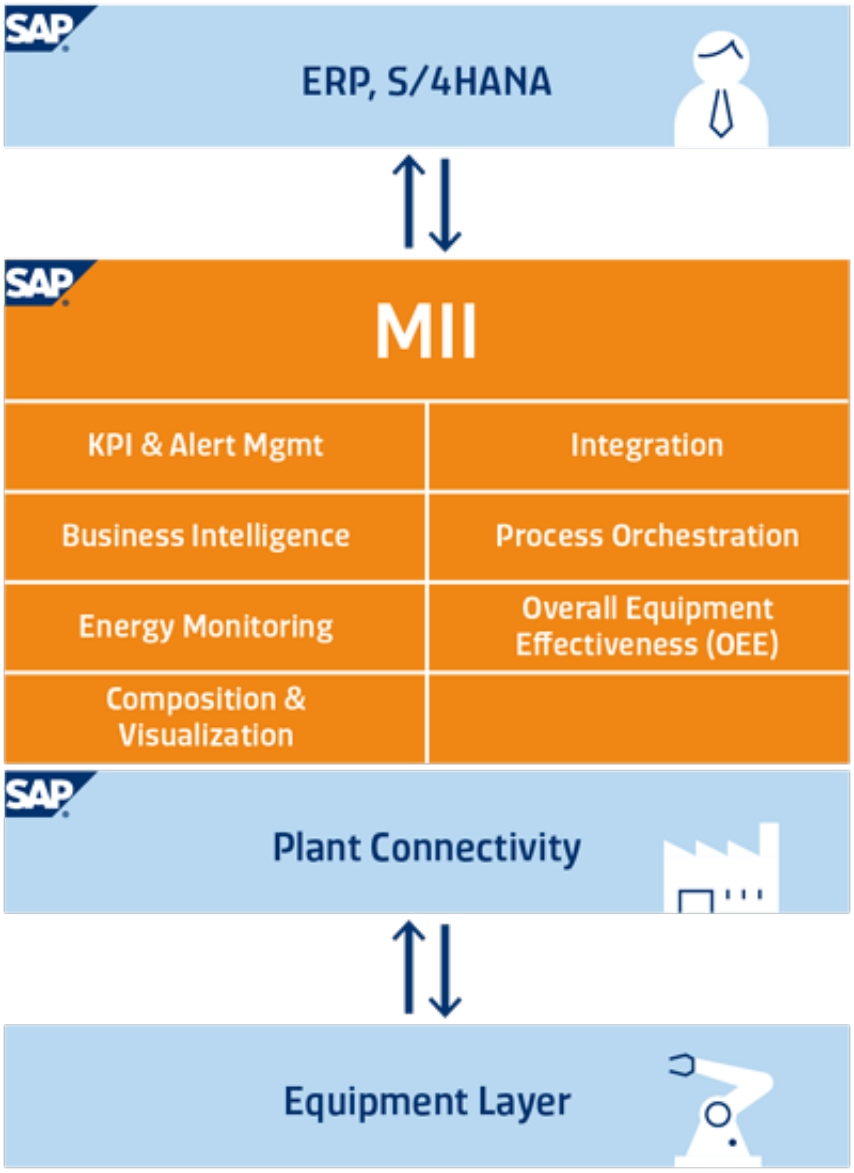


Figure 2.7: Where SAP MII is Located

2.4.2 Benefits of SAP MII

Integrated Systems: Provide interoperability between shop floor solutions and enterprise ERP

Manufacturing Intelligence: With standard, user-defined dashboards, visualize data using plant information appropriate to each level of the business.

Process Orchestration: Enable customer-specific processes for planning, execution, maintenance and quality.

Plant Performance Management: SAP Overall Equipment Effectiveness Management (SAP OEE Management) enables historical and real-time measurement and analysis of plant performance. Options for creating and visualizing customized KPIs are also available.

2.4.3 SAP MII Functional area

MD – Master Data Permits the configuration of complex filling and packaging lines.

- Individual configuration of individual machines and production lines
- Flexible tool for making independent adjustments
- Provides support for special situations such as shared and exclusive resources
- Automatic application of the system configuration to the line control station

SF – Shop Floor Connection Implements the direct inclusion of simple, technical production components.

- Connections to (e.g.) label printers, RFID read/write equipment, scanners and sensor technology (e.g. photoelectric barriers)
- No other IT systems required for connectivity
- Use of homogeneous flow logic for separate production facilities

2.5 SAP Plant Connectivity

2.5.1 briefly

SAP Plant Connectivity (SAP PCo) serves as a link between the equipment and the MES. SAP PCo facilitates data interchange in a manufacturing context between information technology and programmable logic controllers, Industrial Internet of Things (IIoT) devices, plant historian systems, and other devices. PCo requests data from source systems using a variety of protocols. PCo sends

this data to the specified destination system, which might be SAP Manufacturing Integration and Intelligence (SAP MII), SAP Manufacturing Execution (SAP ME), the SAP technology platform SAP HANA, or SAP Extended Warehouse Management (SAP EWM).[**7**]

PCo consists of the following elements:

- Source system
- Target system
- Agent instance
- Notification

SAP Plant Connectivity includes a buffering mechanism for network-based high-speed machine-level communication. This allows resources to be managed so that a business system does not have to be brought up every time.

So to make things clearer, let's have a look at every PCo's element and define exactly its role.

Source System The source system is the system from which PCo retrieves information. A PCo installation can connect with many source systems using manufacturer-independent protocols, such as:

- OLE DB (z. B. ODBC, MS Excel and MS Access)
- File Monitor
- Socket

Target System The target system is the system to which information will be transmitted by PCo. Data can be received by the following target systems:

- SAP Manufacturing Execution (SAP ME)
- SAP Manufacturing Integration and Intelligence (SAP MII)
- SAP Cloud Platform for SAP IoT Solutions
- Business Suite-System, e.g. SAP ERP or SAP EWM

Agent instance SAP PCo is used to establish agent instances that enable high-performance data connections between SAP applications and industry-specific standard sources. Standard sources for the plastics, electronics, and food sectors are a few examples.

Notifications SAP PCo has notification features for monitoring changes to data source components or tags. Messages are launched and the relevant tags are sent to the designated target systems based on the preset triggers. The notification features enhance the operations of the SAP MII Universal Data Server components (UDS).[**7**]

As a way to understand better what we were talking about, let's give an example of a SAP PCo architecture:

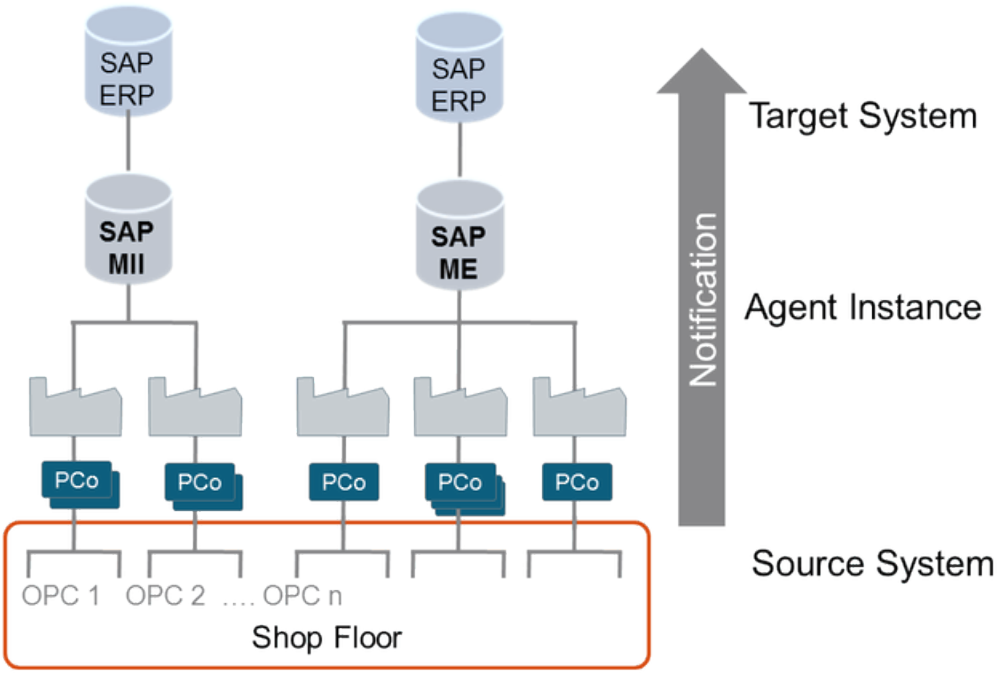


Figure 2.8: SAP PCo Architecture Example

2.5.2 Benefits of SAP PCo

- Rapid Configuration: PCo may be quickly established to connect source systems in production environments without requiring extra development activities inside the current manufacturing IT infrastructure.
- PCo is the sole platform required for equipment integration, allowing communication between equipment and multiple software systems on the shop floor.
- Multi-Site Configuration: A PCo system can be configured for one or more locations

- Extensibility: PCo can be extended via release compatible SDK[**6**]

This chapter defined the most relevant terms and general ideas of this project and helped us make some things clearer so that no difficulties in future use of the terms will be encountered. For the next chapter, we will be discussing the tools needed to make such a project feasible and also we will highlight the conception proposed and the steps that we had to walk through to achieve our goals.

Chapter 3

Application Design and Development

For this chapter, we will at first define the functional and technical needs of our project, then we will give an idea about it by presenting some modeling diagrams and we will finish by giving an idea about the tools required during the implementation of this project.

3.1 Needs Specification

3.1.1 Functional Needs

- A real-time dashboard that allows continuous control of the production chain and the activities within this chain.
- This application allows the prediction of any possible fall or an error that may disrupt the smooth running of activities.
- Makes it easier to non-functional and rather administrative staff to follow unexpected activities and disruptions and therefore ensure the smooth running of the work.
- An interface which is rather user-friendly and which makes it easier for users to control the smooth running of the work.
- Using real-time visualisations of the status of the production process, instant actions can be decided and taken and as a result the losses can be minimized.

3.1.2 Technical Needs

- Easy to understand the dashboard and to identify the exact remarks.

- Operability: the impacts in terms of performance must be taken into account during development like the response time, as well as the consumption of resources (CPU, memory, etc..) which must be minimized.
- Security: each user must authenticate before accessing the application.
- Scalability: the application must always have room for improvement.
- Reliability: the system must operate without failure.

3.2 Needs Modeling

3.2.1 Use Case Diagram

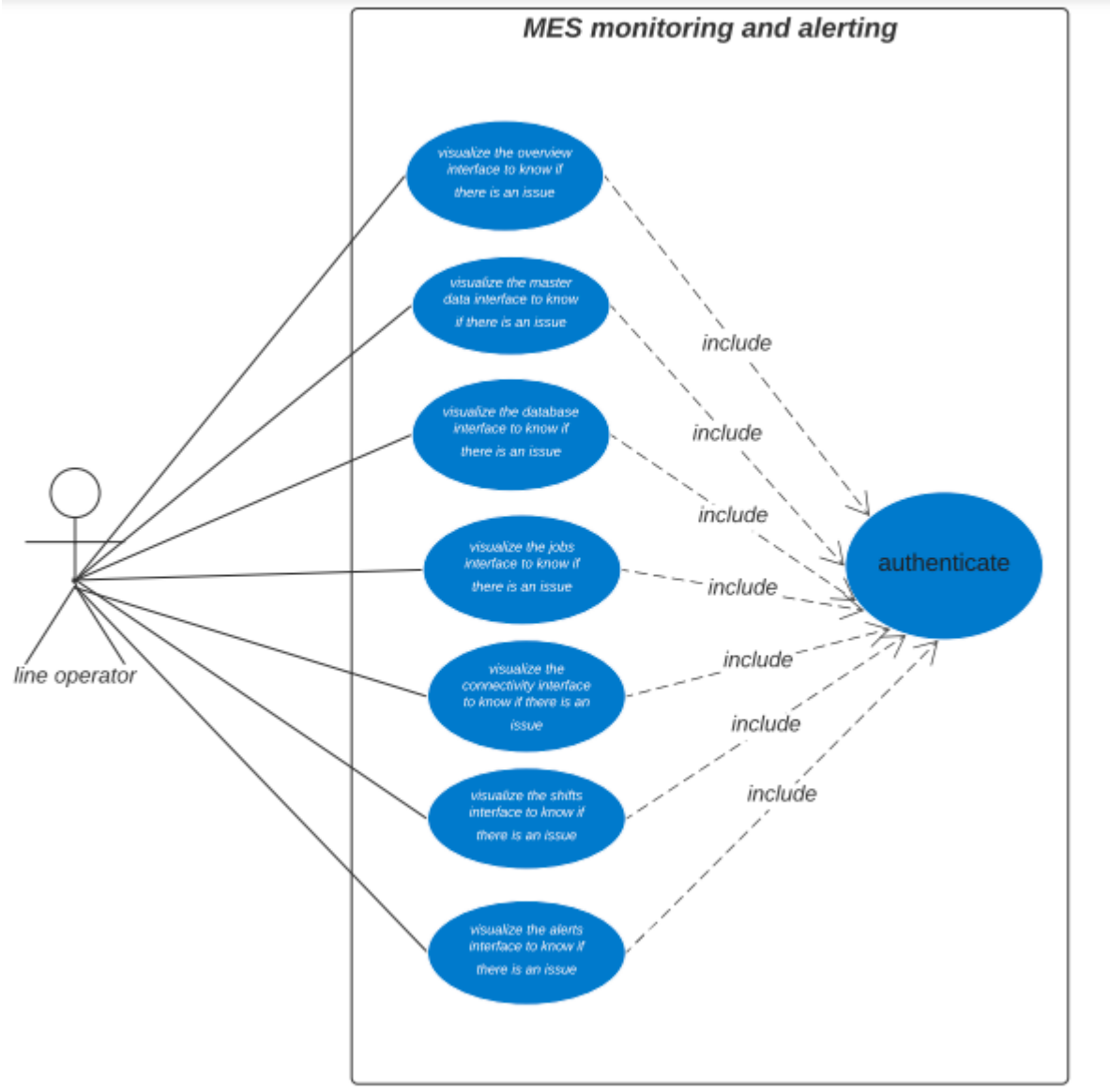


Figure 3.1: Use Case Diagram

3.2.2 Sequence Diagram

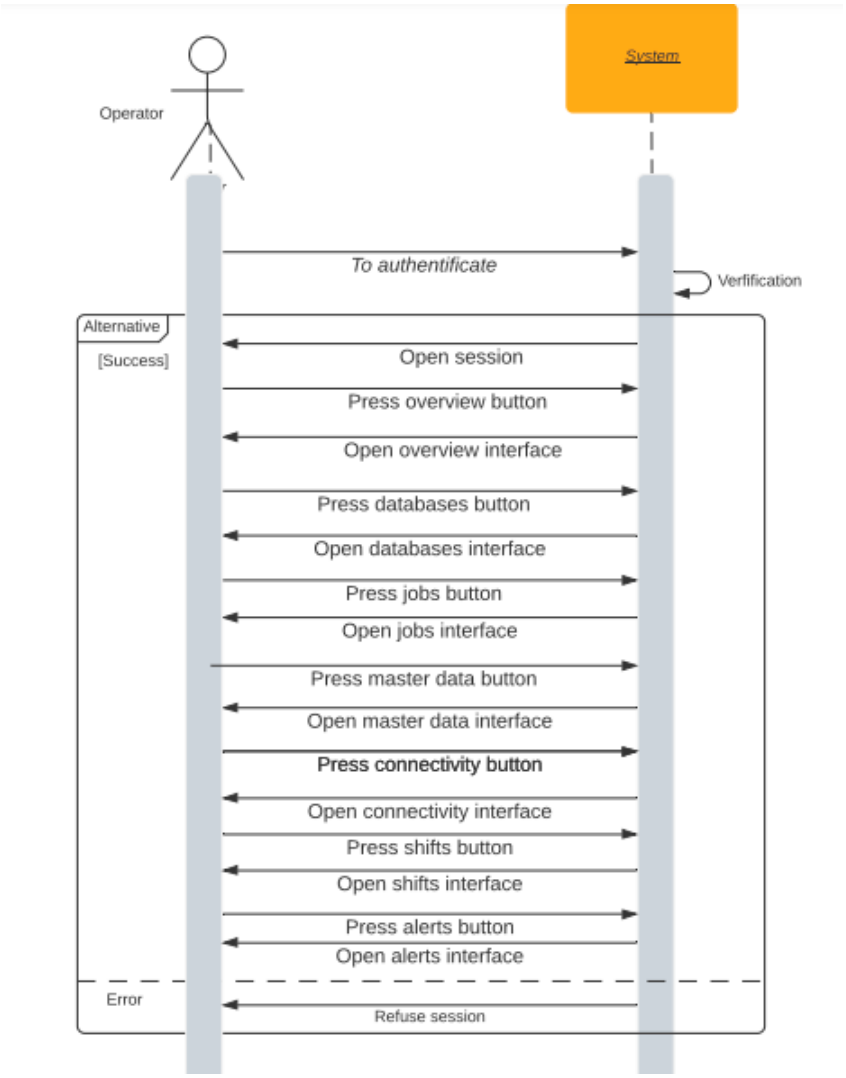


Figure 3.2: Sequence Diagram

3.3 Application Conception

3.3.1 Physical Architecture

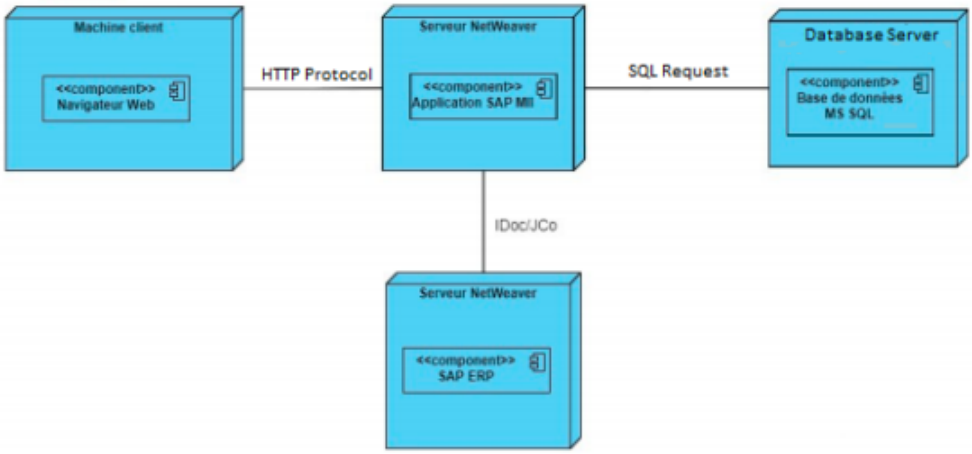


Figure 3.3: Deployment diagram

As shown in the deployment diagram shown in Figure 3.1, the architecture physics of this project is multi-tier (at 4 levels).

There is a level that represents the client machine, a level in which we find the business part of the application, a level which contains the database and a final level containing the SAP ERP software package. The customer level, which corresponds to the visible and interactive part with users, is communicating with the application server using the http protocol. The latter, which represents the functional part of the application, communicates with the data server using SQL queries and with SAP ERP using files of IDoc and JCo to manipulate the data and send the response to the client side.

3.3.2 Logical Architecture

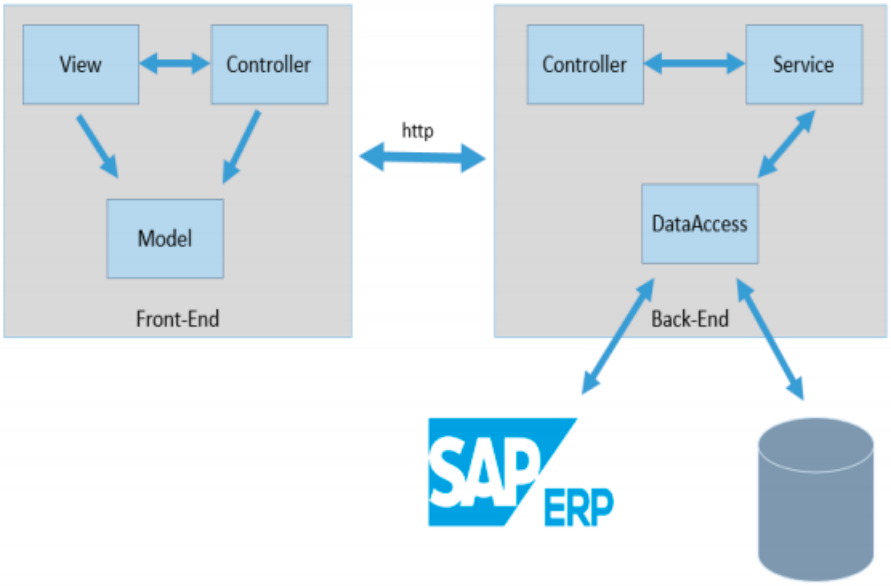


Figure 3.4: Logical Architecture

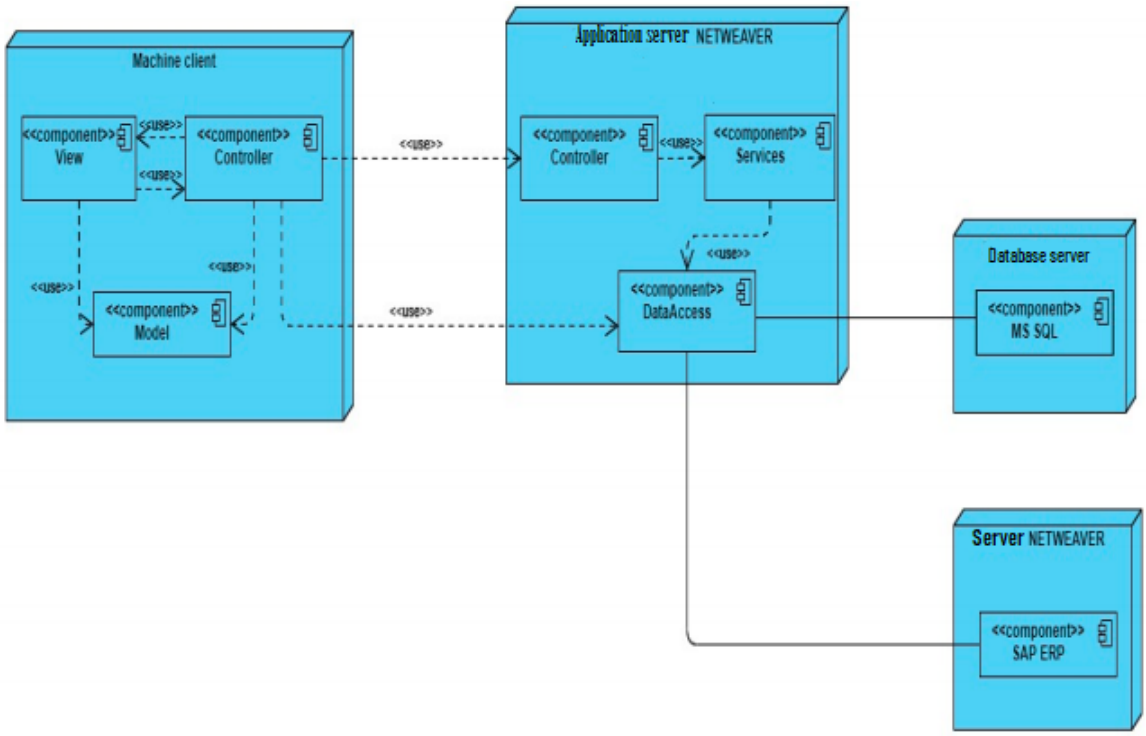


Figure 3.5: component deployment diagram

The logical architecture of our application, as shown in the two figures 3.2 and 3.3, is multi-layered (6 layers). Three of them are in the back-end part and three in the front-end part. Each of these layers has a specific role.

- The View layer: It represents the presentation layer of the application. It interacts with the actions performed by the user by calling the Controller layer and presenting the data returned by the back-end part.
- The Model layer: It contains the data handled by the application.
- The Front-End Controller layer: This layer meets the customer's needs by sending the requests to the back-end part.
- he back-end Controller layer: It acts as an intermediary between the front-end part and the Services layer.
- The Service layer: It contains the business logic of the application.
- The Data Access layer: This layer is responsible for communicating with the database using SQL queries and with SAP ERP using IDocs, BAPIs and RFCs.

3.3.3 Data Model

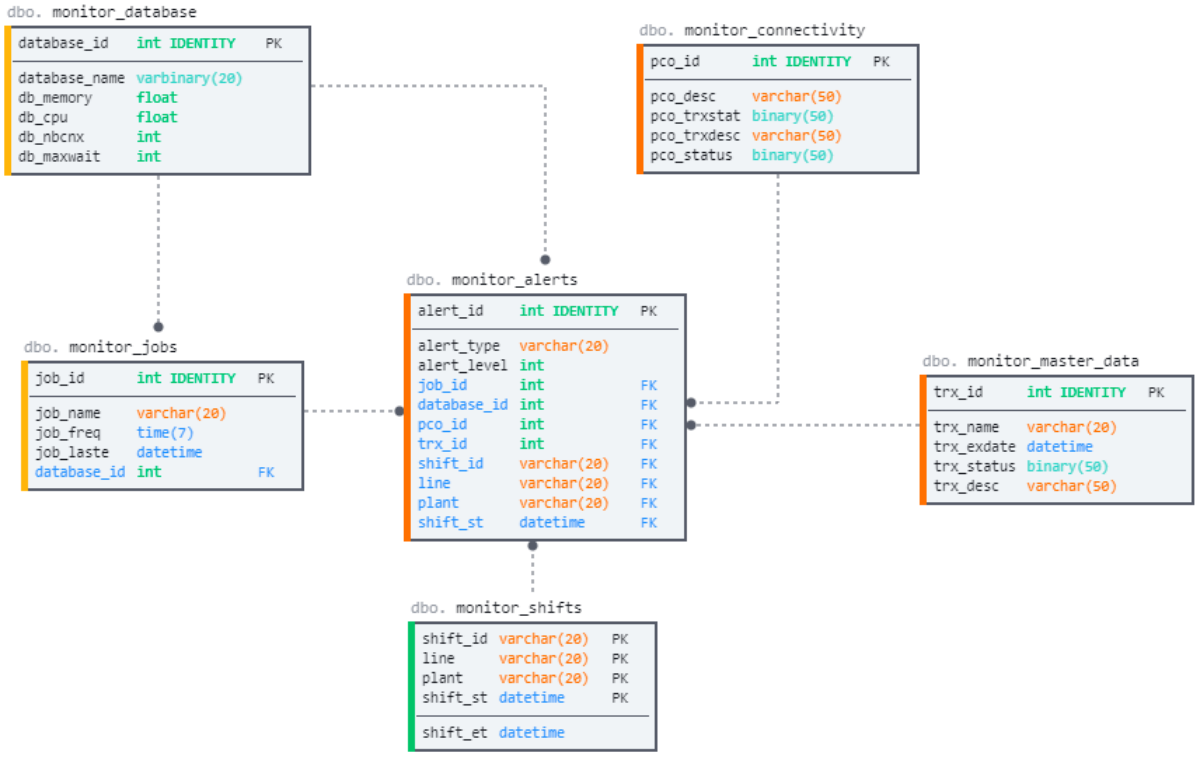


Figure 3.6: Data Model

3.4 Tools Required

For this section, i will be presenting all the tools that i needed during the elaboration of this project either in the back-end development or in the front-end part. And also i will be talking about a MII configuration that needed to be realised to have a clearer look about how things work especially in the back-end part.

3.4.1 Data Collection Tools

MII Installation and Configuration

First of all and before talking about how we used MII during our project, let's talk about an important phase which is the installation and the configuration

of MII.

This step of work was necessary, in fact, we could have use from the start an existing installed and configured MII, but to have a better look at how things really work and especially to be aware about the connection between MII and the database from which data will be loaded which will be SQL in our case, we decided to pass by this step. Briefly, installing and configuring MII from the beginning to the end is the best way to make things clear.

As any installation, we passed by a number of steps :

- Installation of the java development kit (JDK).
- Creation of the transaction in maintenance planner to download a new system.

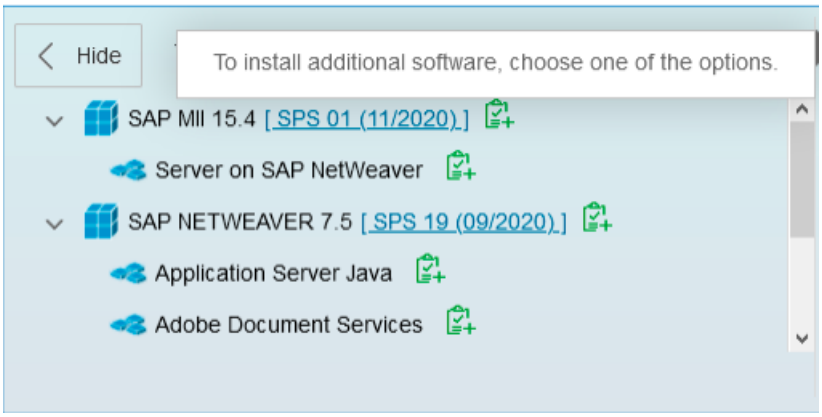


Figure 3.7: Installing MII 15.4 and Netweaver 7.5

During this step we need to install the following components : ADS, Application server Java, Development Infrastructure, Job Scheduler and PDF Export.

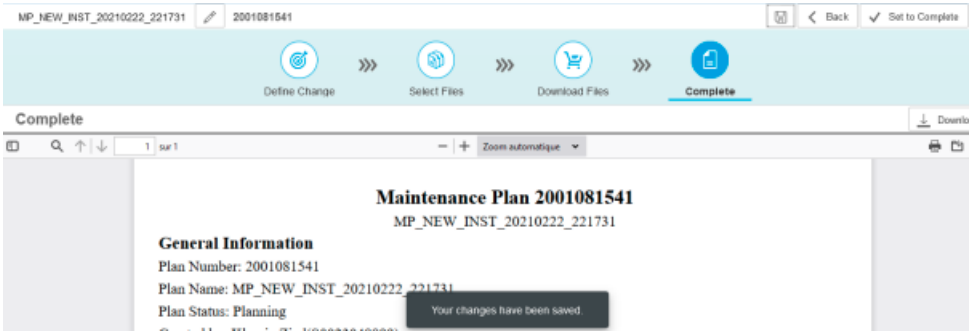


Figure 3.8: Review about the Download Basket

- SQL Server 2012 Installation :

  During this step, we had installed the SQL server that will serve as the database for our MII server.

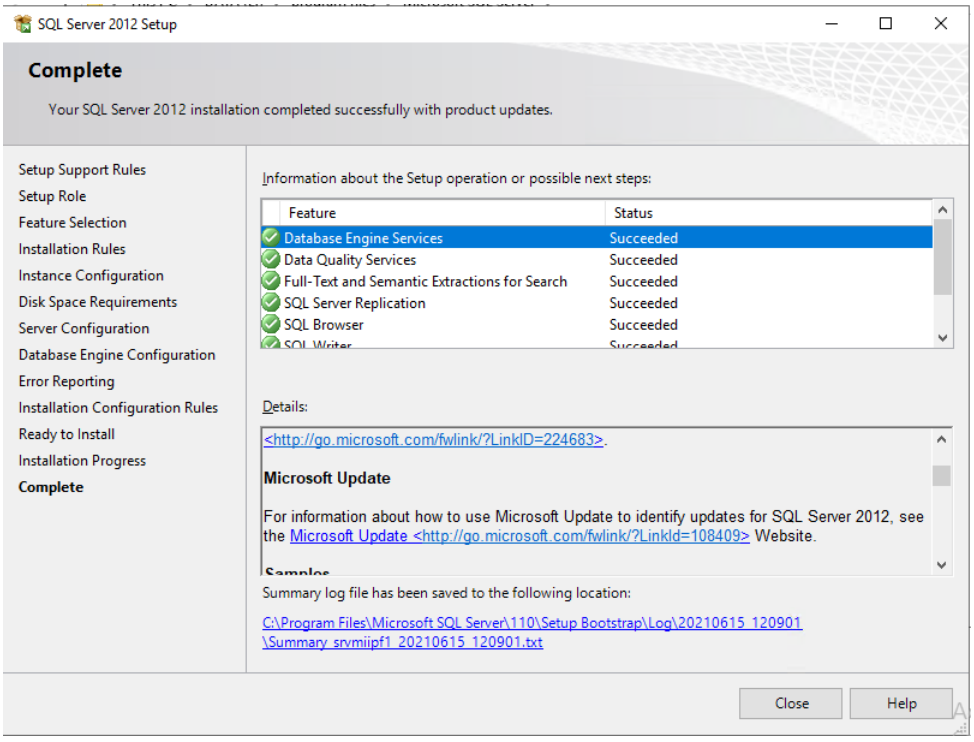


Figure 3.9: SQL Server Installation

- Installation of SQL Studio Management

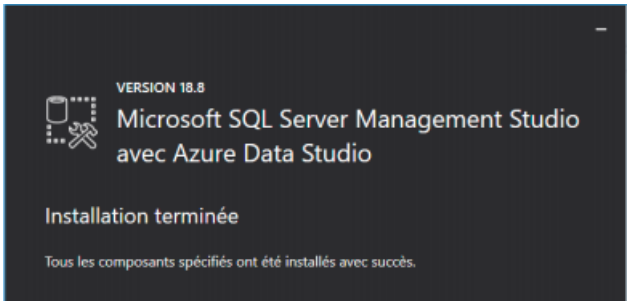


Figure 3.10: SQL server management studio installation

After the installation is complete, we had to achieve a certain step which is the enabling of the sa connection in the SQL server management studio.

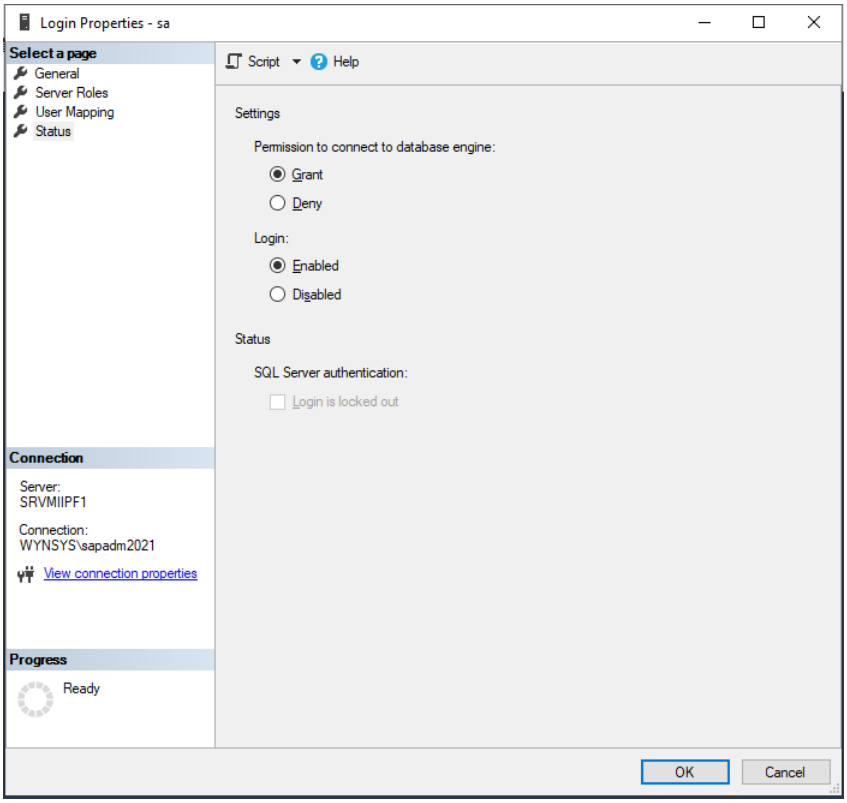


Figure 3.11: Enabling the sa Connection

- Installation of SAP Netweaver : during this step, we will be extracting the setup files using file with the .SAR extension so that we can afterwards launch the installation of the Netweaver to which we will add the ADD ON MII.

```
x SOLMAN72SR2/JAVA/PV/PD/product.xml
x SIGNATURE.SMF
SAPCAR: 854 file(s) extracted
```

Figure 3.12: Extracting files

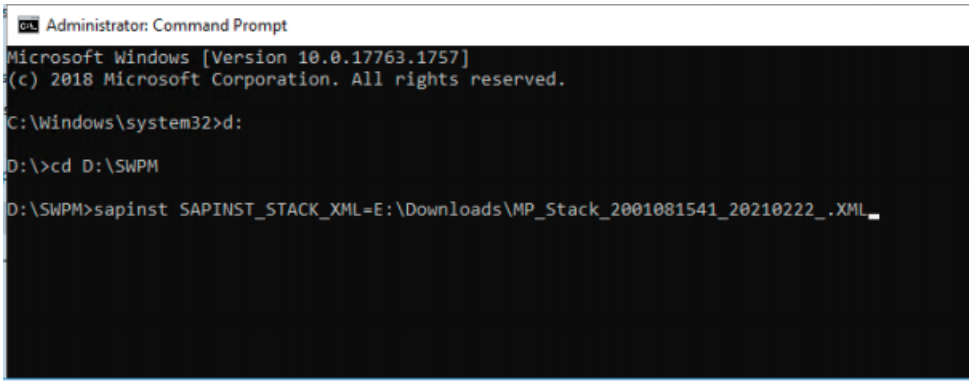


Figure 3.13: Launching the installation with SWPM

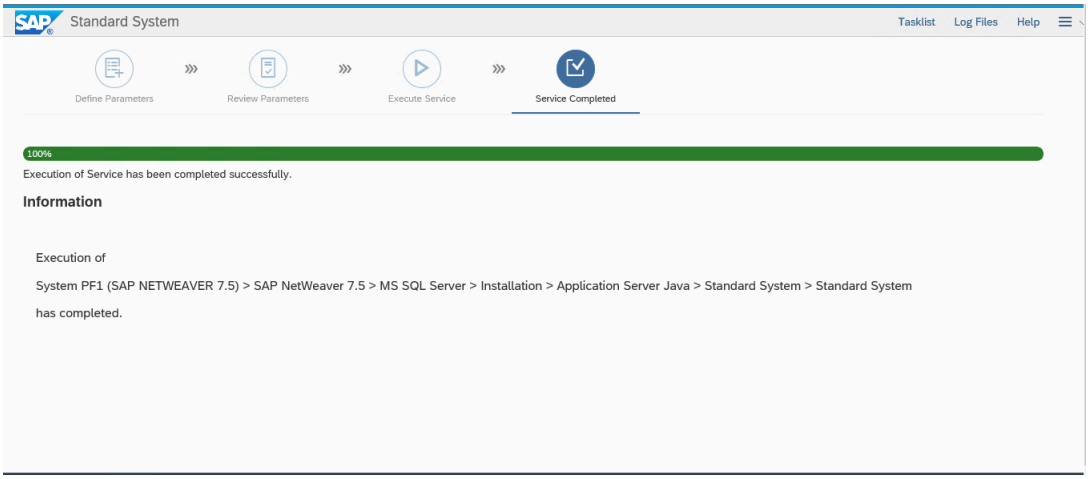


Figure 3.14: Installed Successfully

After ensuring that the installation was successfully done, we can open the Netweaver that we just installed and have a look at the process and also ensuring that they are all running.

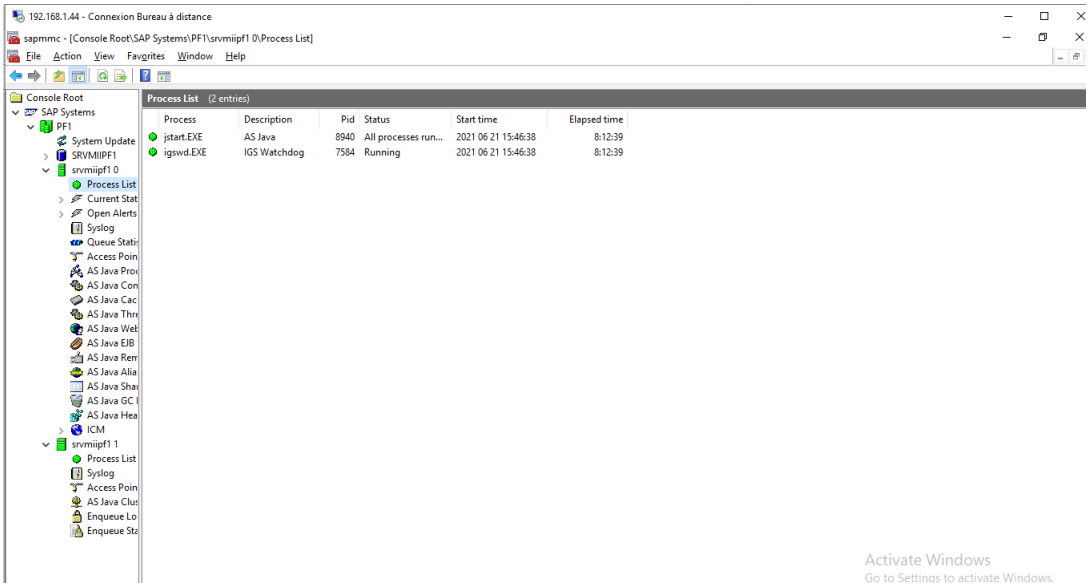


Figure 3.15: Process Verfication

- Installation of ADD ON MII : As we mentioned earlier, we need to install the ADD ON MII on the Netweaver that we recently installed. As a result of that we will be able to create our MII server within this Netweaver.

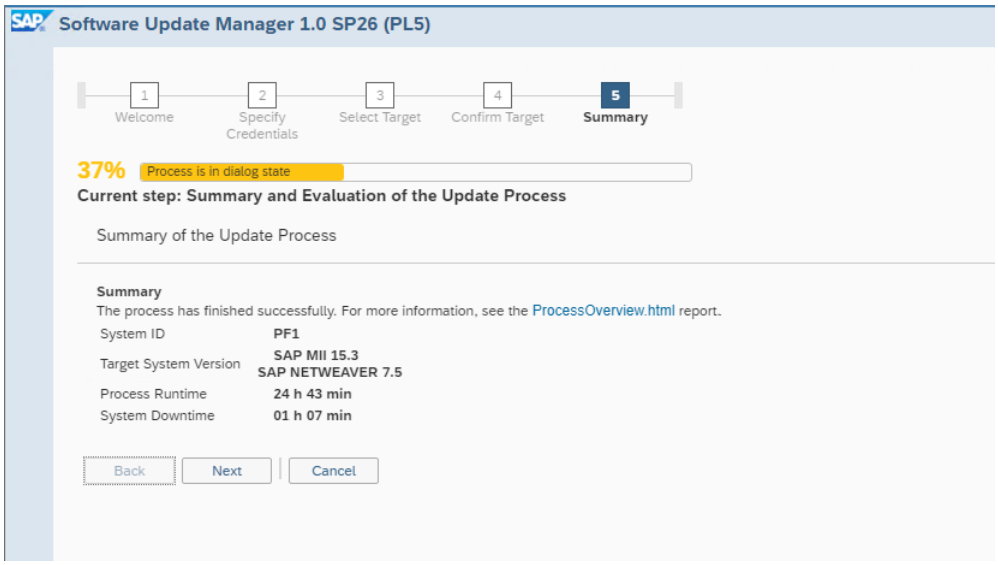


Figure 3.16: Summary of the Process

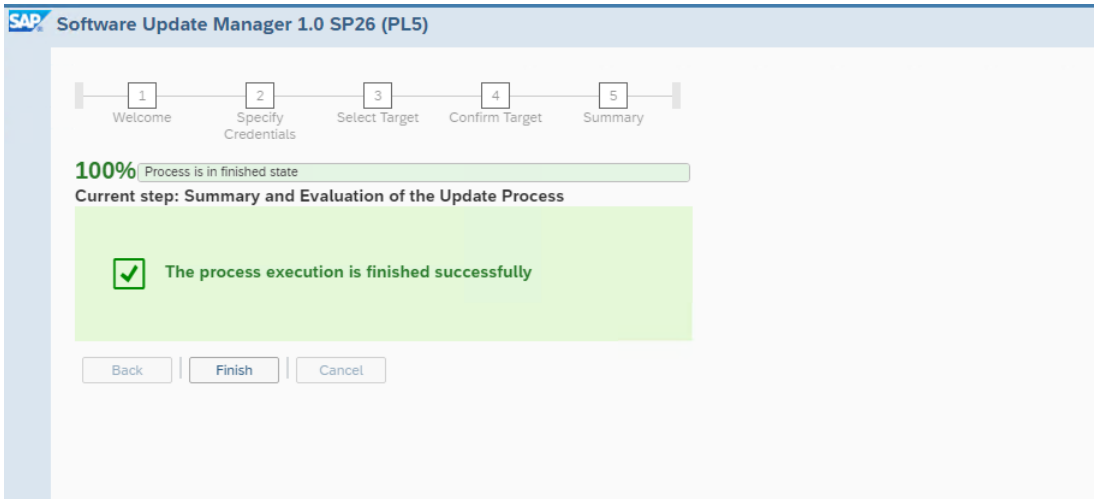


Figure 3.17: MII Installed Successfully

- Creation of the SQL Server connection to establish the connection between MII and the operational database :

Creation of the Operational MII Database

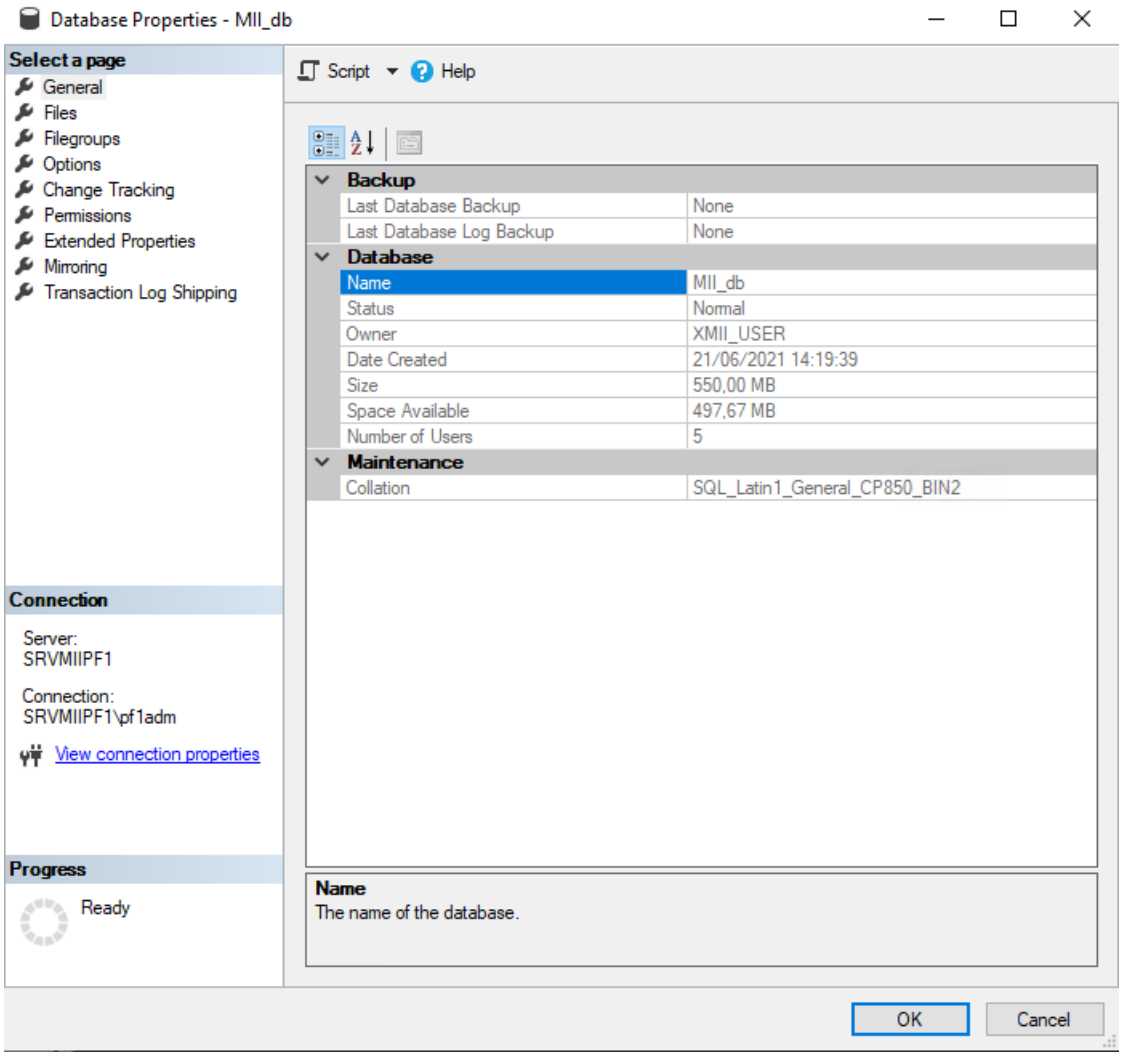


Figure 3.18: Creating the Database

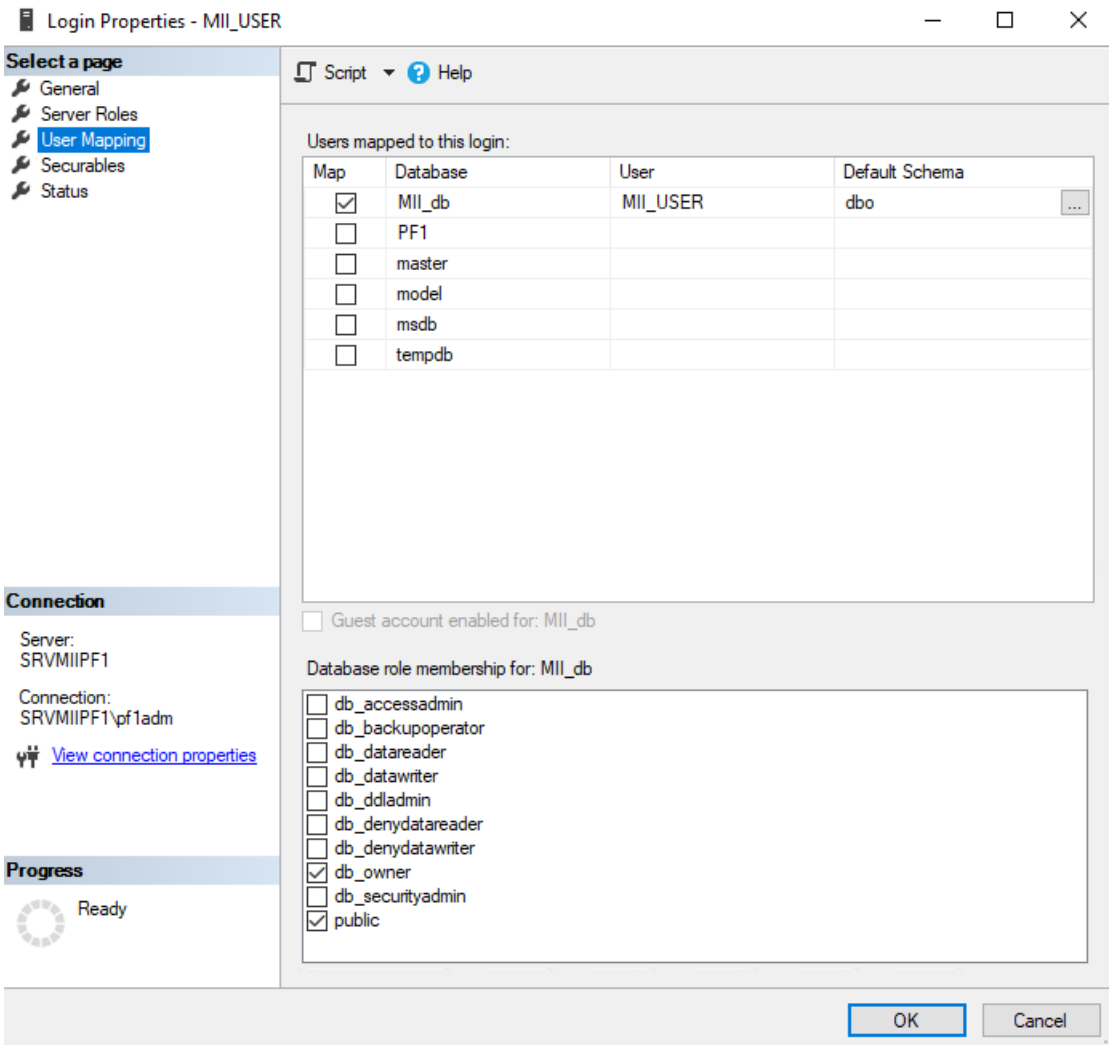


Figure 3.19: Creating the user and Linking it to the Database

Creating the connection between MII and the database First of all, we nned to create a new MII server that we will call DATABASE-MII and link it to the SQL database that we already created.

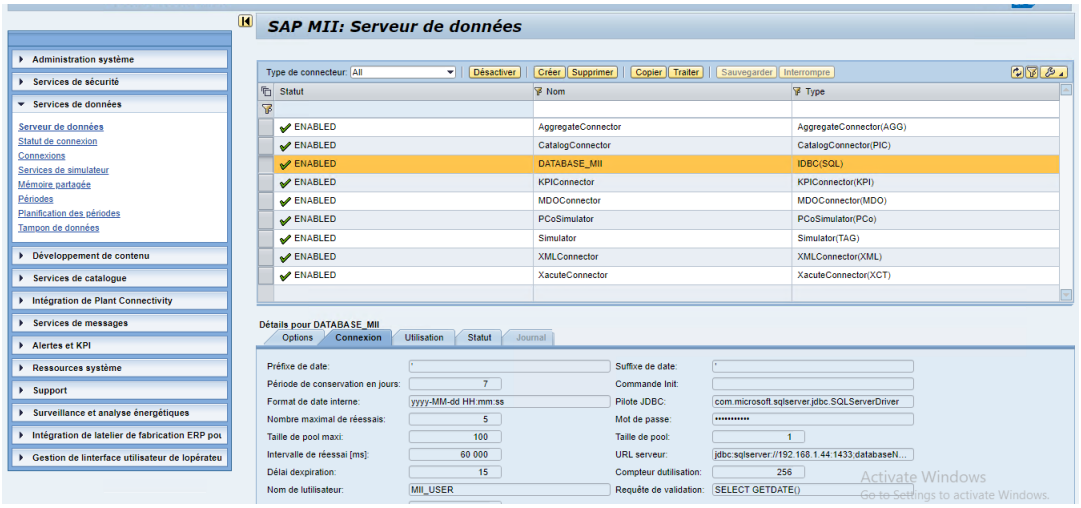


Figure 3.20: Creating the New Server

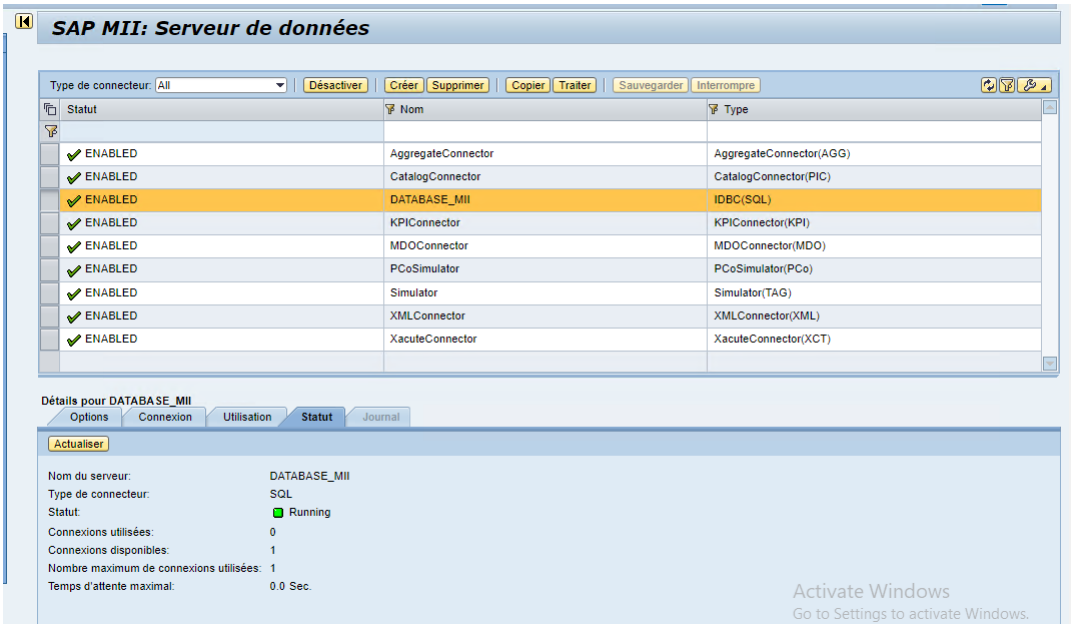


Figure 3.21: Verifying the Status of the Connection

So as a conclusion to all this part of installations and configurations, we were able to successfully create a MII server that his connection to the SQL

server was established with no issues and as a result we can now use our proper MII server that its database is SQL as we have chosen from the start and also elaborate our back-end part in our proper server.

SAP MII Workbench

SAP MII Workbench is the environment that facilitates the development of queries and transactions. It allows us to collect and manipulate data from several sources using BLS (Business Logic Services). BLS allows us to develop a transaction in the form of a graph. A transaction is executed from top to bottom and left to right and is triggered via a scheduler (Job) or by triggering an event on a web page. A transaction consists of a set of interdependent sequences. Each sequence contains one or more actions. An action performs a specific task in a transaction. We can find the action conditional which decides which schema will be executed, Repeater which has a function similar to the For loop, Switch, etc. An action can contain a request, perform a transformation or a data calculation, contain messaging services, etc.

Integration with SAP ERP Communication between SAP ERP and SAP MII is provided in different ways. The IDocs and Java Connector (JCo) files are frequently used to provide this link.

IDocs Files : An IDoc, which stands for an intermediate document, is a container data that can be used to exchange information between two SAP systems or between an SAP system and an external system capable of interpreting this type of file.

JAVA Connector (JCo) : JCo is an API programming interface that allows to establish the connection between SAP MII (Java languages in general) and SAP ERP (languages ABAP) to execute BAPIs (functions) and RFCs (query a table). SAP MII offers a set of actions that are usually executed in the following order :

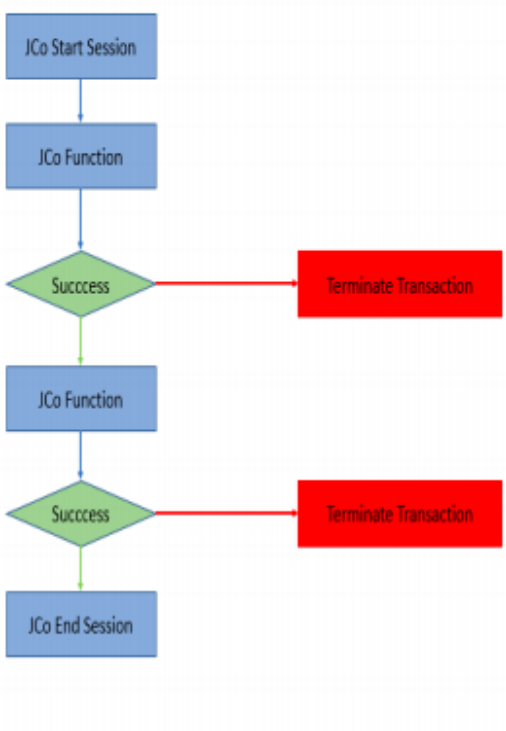


Figure 3.22: The JCo actions execution scheme

The actions used are: SAP JCo Start Session, SAP JCo Function, Conditional, SAP JCO End Session and Terminate Transaction.

3.4.2 Interfaces Development Tools

SAPUI5

SAP UI5 is the latest user interface development technology from SAP, launched in 2013.

Although it was originally designed to create SAP applications, it is now used for the development of all web applications.

SAP UI5 is a framework that contains a set of libraries, is based on HTML5, CSS and Javascript and allows you to create responsive web applications, which means interfaces automatically adapt to the device on which the application is running.

SAP UI5 offers translation support. Indeed, the interfaces are automatically translated into the language of the browser.

SAP UI5 is based on the MVC architectural concept to create a separation between the business logic, data and views.[**8**]



Figure 3.23: SAPUI5

SAP Business Application Studio

SAP Business Application Studio is a new SAP Business Technology Platform (BTP) service that provides a contemporary development environment optimized for the effective creation of SAP Intelligent Enterprise business applications.

SAP Business Application Studio, which is available as a cloud service, delivers a desktop-like experience similarly to popular IDEs, complete with command line and optimized editors. The dev spaces are at the heart of SAP Business Application Studio. The dev spaces are similar to separate virtual computers in the cloud that feature specialized tools and preloaded run-times based on the business situation, such as SAP Fiori, SAP S/4HANA extensions, Workflow, Mobile, and others. This simplifies and saves time in setting up your development environment, allowing you to create more efficiently.[**9**]

Conclusion

All along this chapter, we tried to cover the most important parts about the development of our application beginning by the specification of the needs either functional or technical and also modeling those needs in some useful diagrams, passing by presenting the application conception and especially the data model and concluding at the end by defining the tools needed during the realisation of the project and also some of the installations and configurations that we

estimated essential to talk about. The next and last chapter will be dealing with the realisation of this application and the results obtained at the end of this project and some discussion about the efficiency of this results.

Chapter 4

Realisations and Results

For this chapter, we will be presenting all the implementations and the results achieved during this project and also before that we will talk a little bit about the work environment.

4.1 Work Environment

4.1.1 Hardware Environment

When carrying out this project, we are using a "Lenovo ThinkPad" PC with the following characteristics:
- The processor: Intel Core i5
- The hard drive: 1 TB.
- RAM memory: 8 GB.
- The operating system: Windows 10.

4.1.2 Software Environment

SAP Cloud Platform

SAP Cloud Platform is a platform hosted in the cloud in PaaS mode (Platform as a service) and developed by SAP. It offers many services such as the Internet of objects (SAP Cloud Platform Internet of Things), analysis services (SAP Cloud Platform Predictive services, etc.), integration services (SAP Cloud Platform Predictive services, etc.). The service used in this project is SAP Cloud Platform Web IDE. It's an editor for develop web applications based on SAPUI5.

Manufacturing Integration and Intelligence (SAP MII)

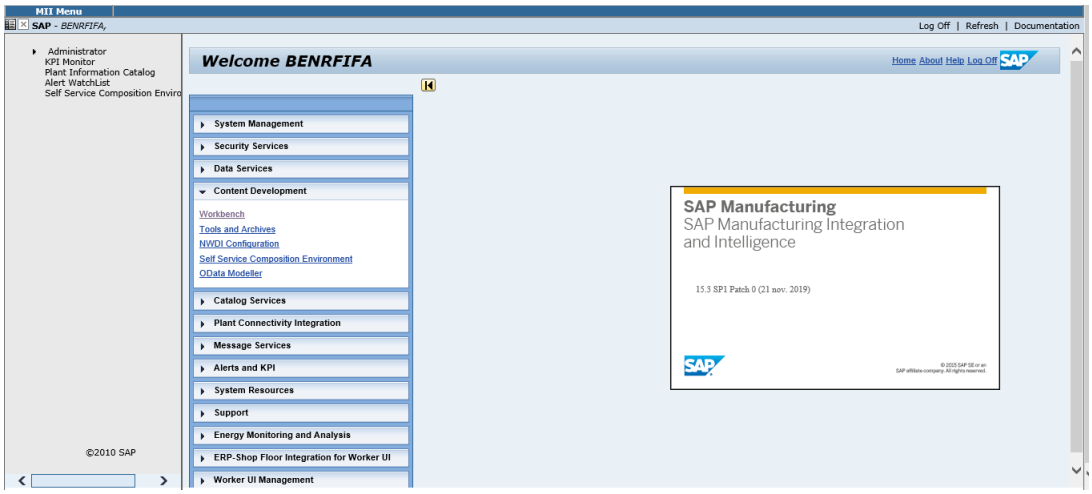


Figure 4.1: Manufacturing Integration and Intelligence

SAP MII is a platform used in the industrial sector, the main objective of which is is to provide data integration and analysis. SAP MII enables visualization alerts and KPIs using visualization tools such as charts and tables on board. In addition, it allows integration with SAP ERP in real time, which ensures the synchronization between the company headquarters and the factory. This integration is done using JCo IDocs and API files.

SAP MII offers developers an editor called SAP MII Workbench which allows the development of different algorithms based on Business Logic Services (BLS) and therefore on graphs.

Microsoft SQL Server (MS SQL)

Microsoft SQL Server is a database management system (DBMS) language SQL incorporating among others an RDBMS (relational DBMS) developed and marketed by the Microsoft company.

4.2 Implementations

For this section, we will be dealing with the results achieved at the end of this project.

4.2.1 Data Collection

Data from the MII

Creating the Data Tables First of all, we have to begin by creating the tables that will store our data and therefore be the source from which our interfaces in the front-end part will take their input.

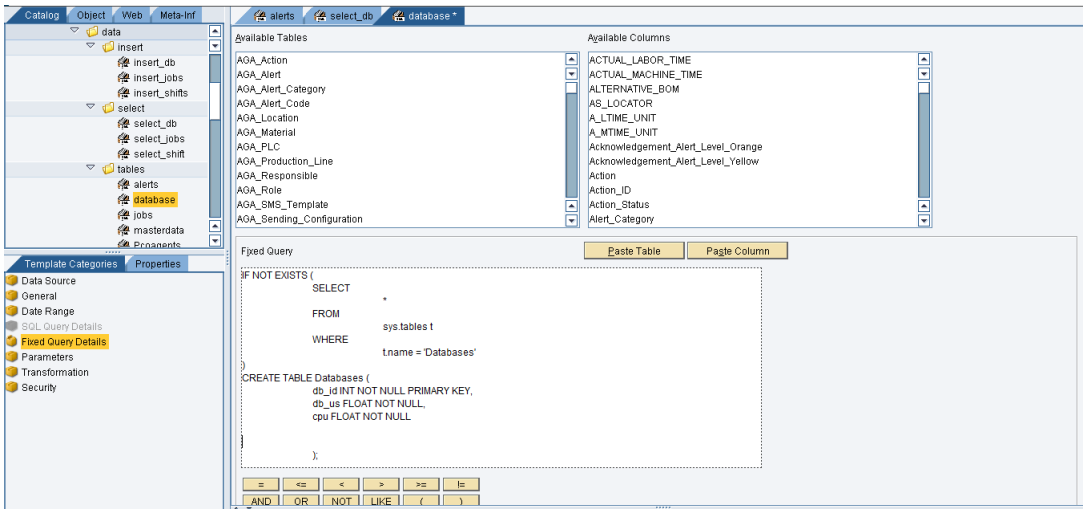


Figure 4.2: Creating the Database Table

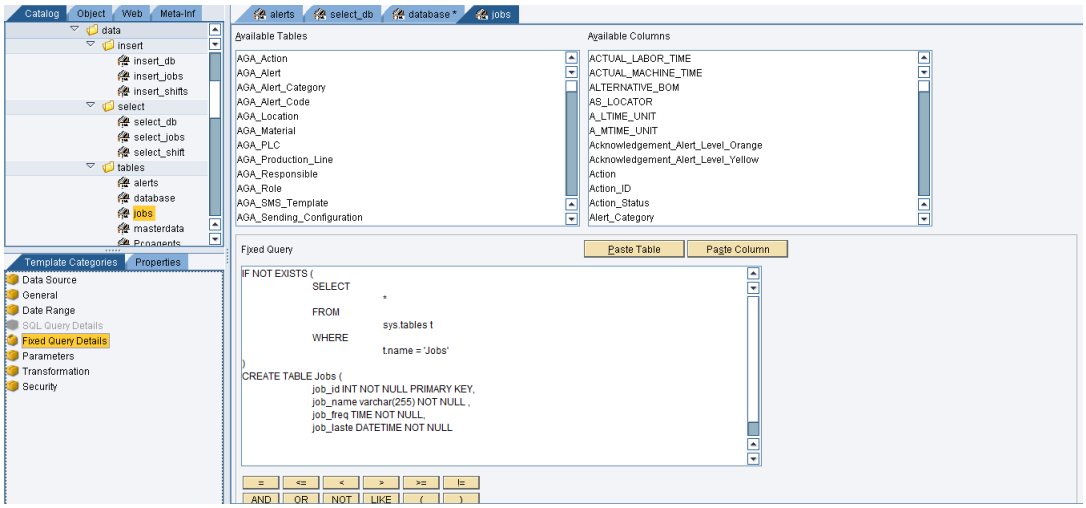


Figure 4.3: Creating the Jobs Table

So as the 4.2 and 4.3 figures show, we created the tables of databases and jobs with the needed attributes that we will need afterwards in our interfaces.

Data Collection For this step, we will be collecting the data and putting it in the Table in question.

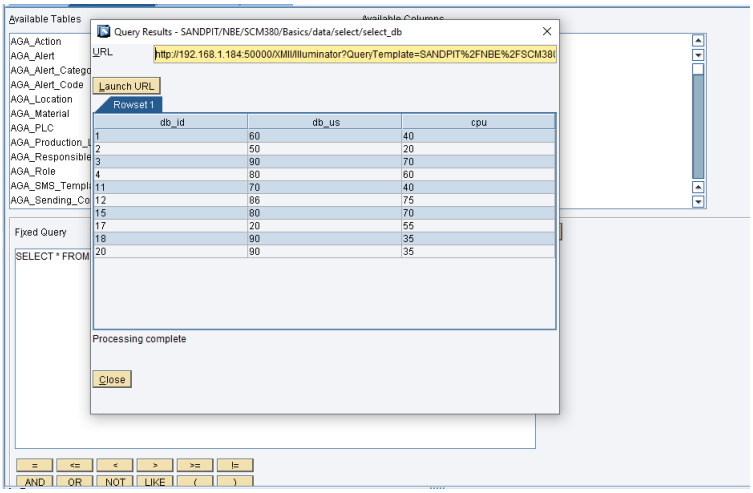


Figure 4.4: Database Table

```
{ "Rowsets" : {
        "DateCreated" : "2021-06-24T13:26:05", "Version" : "15.3 SP1 Patch 0 (21 nov. 2019)", "StartDate" : "2021-06-24T12:26:05+0100", "EndDate" : "2021-06-24T13:26:05+0100", "CachedTime"
: "",
        "Rowset" : [
        { "Columns" : {
                "Column" : [
                { "Name" : "db_id", "SourceColumn" : "db_id", "Description" : "db_id", "SQLDataType" : 4, "MinRange" : 1.0, "MaxRange" : 1.0 },
                { "Name" : "db_us", "SourceColumn" : "db_us", "Description" : "db_us", "SQLDataType" : 8, "MinRange" : 1.0, "MaxRange" : 1.0 },
                { "Name" : "cpu", "SourceColumn" : "cpu", "Description" : "cpu", "SQLDataType" : 8, "MinRange" : 1.0, "MaxRange" : 1.0 } ] },
        "Row" : [
                { "db_id" : 1, "db_us" : 60.00, "cpu" : 40.00 },
                { "db_id" : 2, "db_us" : 50.00, "cpu" : 20.00 },
                { "db_id" : 3, "db_us" : 90.00, "cpu" : 70.00 },
                { "db_id" : 4, "db_us" : 80.00, "cpu" : 60.00 },
                { "db_id" : 11, "db_us" : 70.00, "cpu" : 40.00 },
                { "db_id" : 12, "db_us" : 86.00, "cpu" : 75.00 },
                { "db_id" : 15, "db_us" : 80.00, "cpu" : 70.00 },
                { "db_id" : 17, "db_us" : 20.00, "cpu" : 55.00 },
                { "db_id" : 18, "db_us" : 90.00, "cpu" : 35.00 },
                { "db_id" : 20, "db_us" : 90.00, "cpu" : 35.00 } ]
                 } ]
         }
 }
```


Figure 4.5: Data in JSON Format

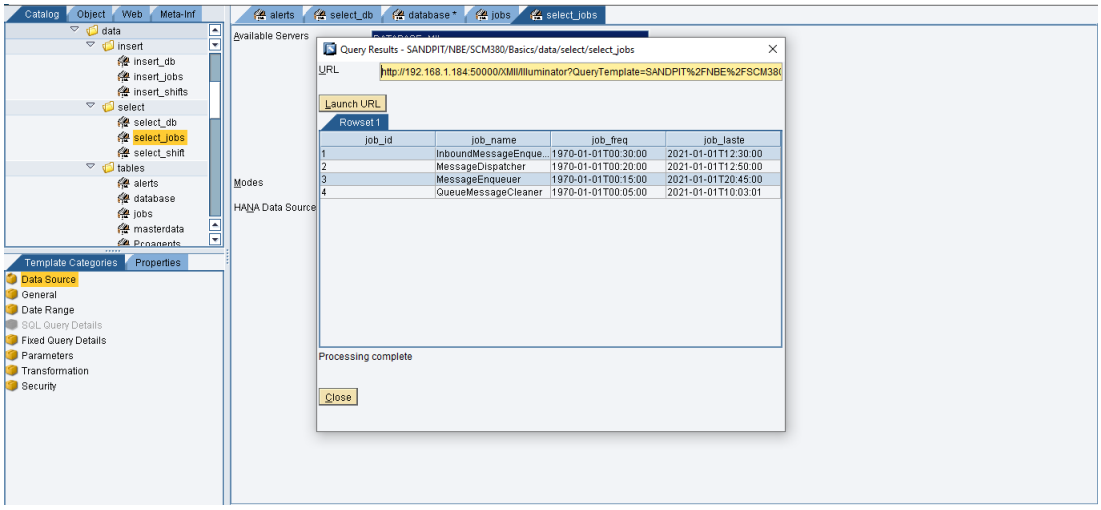


Figure 4.6: Jobs Table


```
{ "Rowsets" : {
        "DateCreated" : "2021-06-24T13:37:16", "Version" : "15.3 SP1 Patch 0 (21 nov. 2019)", "StartDate" : "2021-06-24T12:37:16+0100", "EndDate" : "2021-06-24T13:37:16+0100", "CachedTime"
: "",
        "Rowset" : [
        { "Columns" : {
                "Column" : [
                { "Name" : "job_id", "SourceColumn" : "job_id", "Description" : "job_id", "SQLDataType" : 4, "MinRange" : 1.0, "MaxRange" : 1.0 },
                { "Name" : "job_name", "SourceColumn" : "job_name", "Description" : "job_name", "SQLDataType" : 12, "MinRange" : 1.0, "MaxRange" : 1.0 },
                { "Name" : "job_freq", "SourceColumn" : "job_freq", "Description" : "job_freq", "SQLDataType" : 92, "MinRange" : 1.0, "MaxRange" : 1.0 },
                { "Name" : "job_laste", "SourceColumn" : "job_laste", "Description" : "job_laste", "SQLDataType" : 93, "MinRange" : 1.0, "MaxRange" : 1.0 } ] },
        "Row" : [
                { "job_id" : 1, "job_name" : "InboundMessageEnqueuer_XMIIIDOC01", "job_freq" : "01/01/1970 00:30:00", "job_laste" : "01/01/2021 12:30:00" },
                { "job_id" : 2, "job_name" : "MessageDispatcher", "job_freq" : "01/01/1970 00:20:00", "job_laste" : "01/01/2021 12:50:00" },
                { "job_id" : 3, "job_name" : "MessageEnqueuer", "job_freq" : "01/01/1970 00:15:00", "job_laste" : "01/01/2021 20:45:00" },
                { "job_id" : 4, "job_name" : "QueueMessageCleaner", "job_freq" : "01/01/1970 00:05:00", "job_laste" : "01/01/2021 10:03:01" } ]
                 } ]
         }
 }
```


Figure 4.7: Jobs Data in JSON Format

So as we can see in the 4.4 figure, we collected some data and stored it in the database table and also we can put the data in the JSON format as the figure 4.5 show and so we will be able to deal with this data in the front-end part. And the same thing with the Jobs table in the 4.6 and 4.7 figures. As for the mechanism of importing data, we used some web services called illuminator services of SAP MII that allow us to fetch data from MII and so use it in the front-end part. Those services allow us to :

- Only way to fetch data from MII internal Table
- Performance wise one of the best way to use that in the code
- Easy to use directly from the IRPT or from JS

Background of MII services: MII Services or illuminator services are basically a set of HTTP services that provide some core functionality of SAP MII. We can call these services externally by XML message exchange.

Each illuminator services have a service name and one or more modes which are kind of operations of the service. At the time of execution, we must have to specify both the service name and mode with the input parameter if any. The services can be called when programming or from the web pages by using AJAX.
The pattern of the URL is:
http://*server*:*port*/XMII/Illuminator?service=*servicename*&*mode* =< *modename* >
&content-type=text/xml

From this URL, we can choose the server and the port from where we are importing data and also the name of the service that we need.

Data Collection from PCo

For this part, as we collected data from MII in the last section, we will be collecting data from PCo but we will as well import it to the MII to be able to deal with it later as the other data.

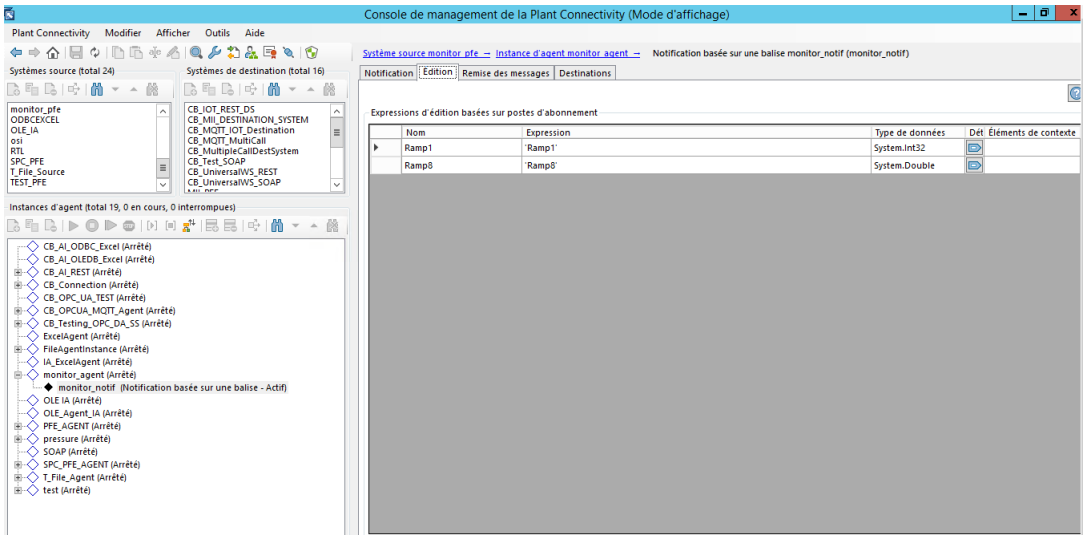


Figure 4.8: Creating the PCo agent in the PCo server

So, as the figure 4.8 shows, we have first of all to create the PCo agent in the PCo server from where we are importing data and then link this agent with the MII server where our data will be stored.

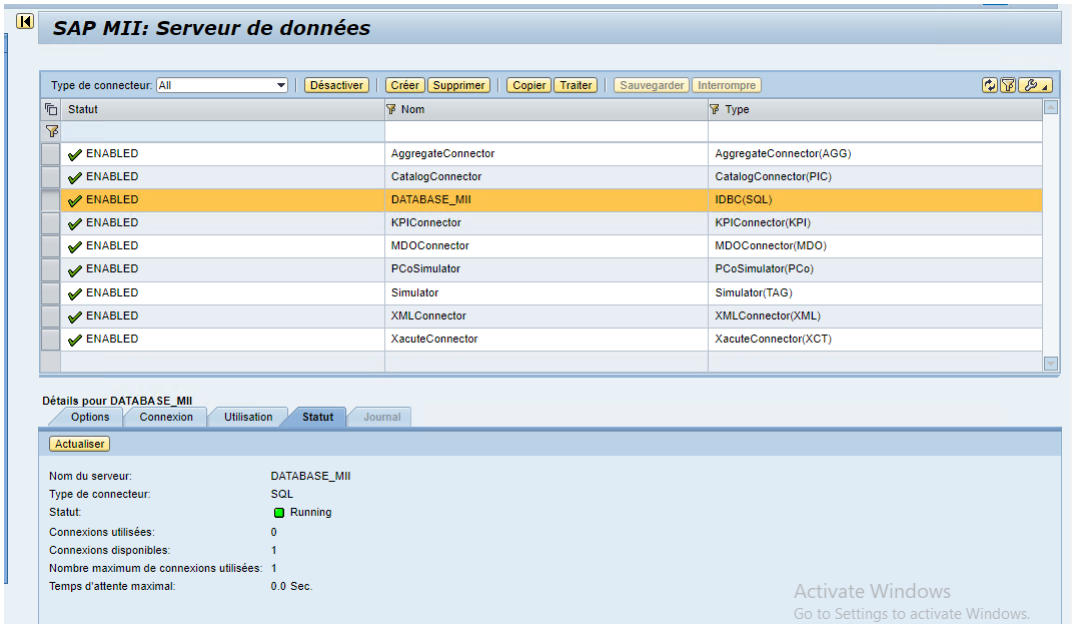


Figure 4.9: Creating the Data Server

For this step, we have to create a data server of a PCo type that we will after that extract the status of this data server using like for the paragraph before, some illuminator services. This services will give us information about all the data servers and we have to filter those results following the type either SQL for the databases in MII or PCo for the data from PCo.

4.2.2 Application Interfaces

Overview

So for this interface, it will be a summary of all the other interfaces and it will display the general status of the other interfaces.

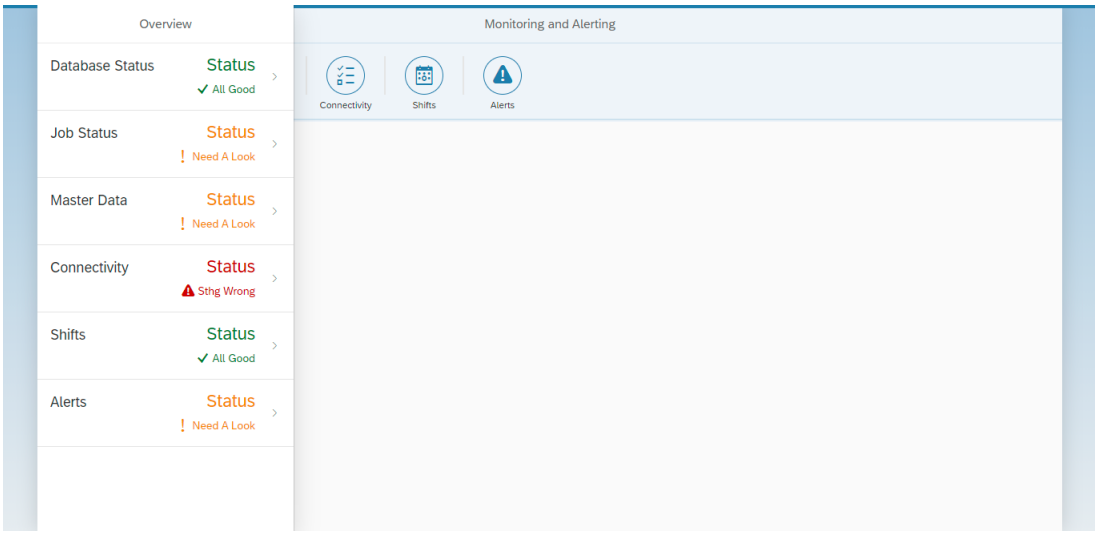


Figure 4.10: Overview Interface

Databases

For this interface, it will display important information about the databases included in the work, some of this information may be the used and so the free memory in this database, how much this database is consuming in term of CPU and also an important information that can be displayed is haw much connections are established with this database which can give us an idea about the flow of the work and also the time that we had to wait until the request is fulfilled.

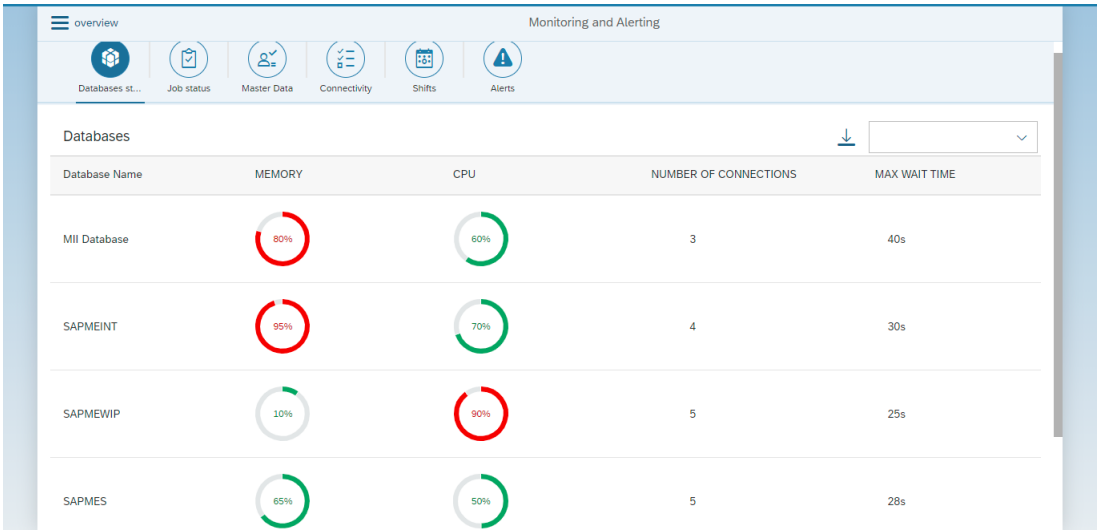


Figure 4.11: Databases Interface

Jobs

Concerning this interface, it will be dealing with the jobs that must be run every a certain period of time, as a result we had to make sure that this job is running in the exact time and more importantly it's not taking much time than usual. The jobs in this interface may be numerous, for that we need a filter to pick the needed job that we want to have a look at and that's the idea deployed in the figure 4.13. We can also view the log of any job by clicking the button details as the figure 4.14 shows.

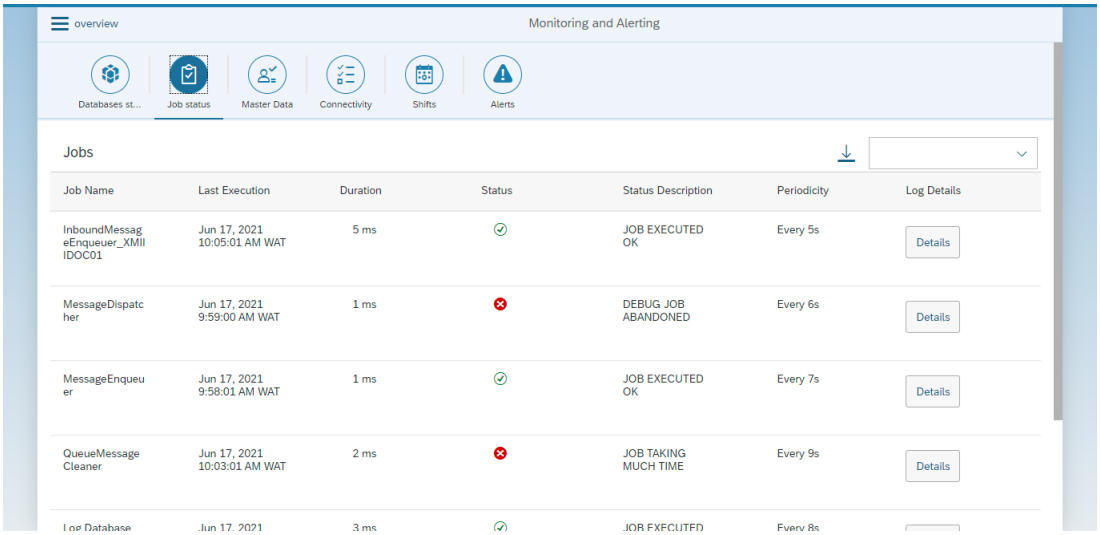


Figure 4.12: Jobs Interface

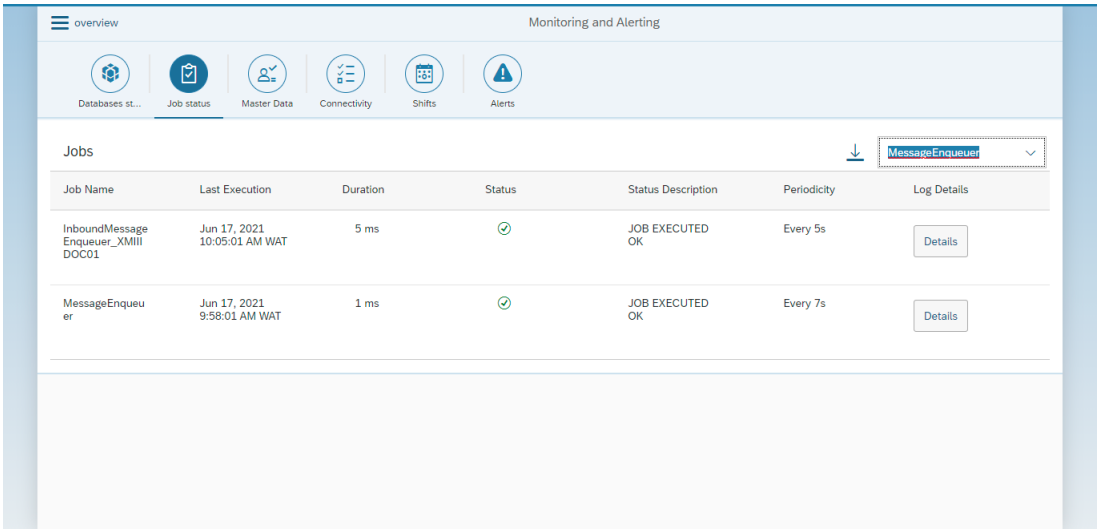


Figure 4.13: Jobs Interface with Filter

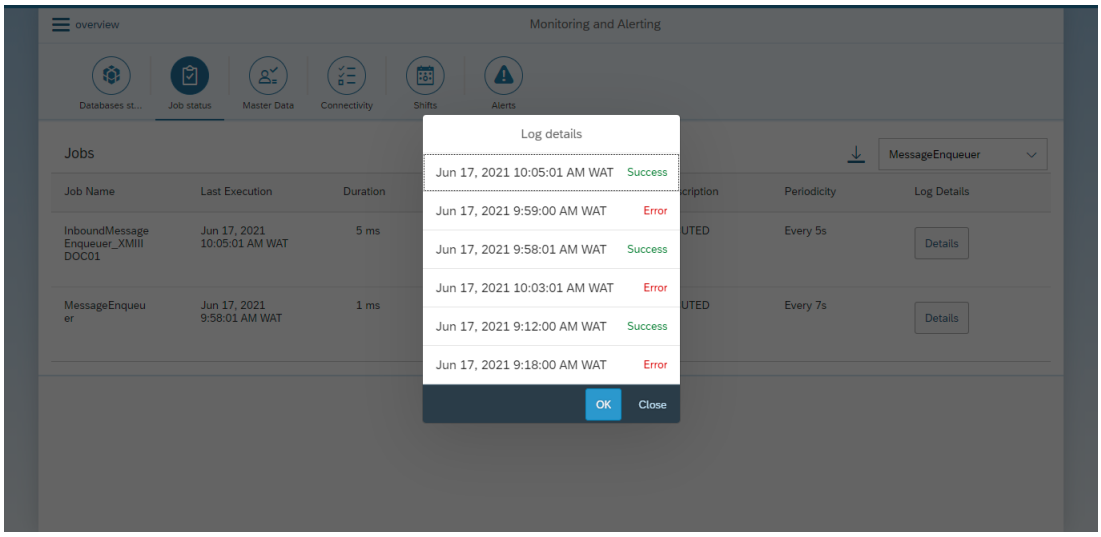


Figure 4.14: Jobs Interface : Job details

Master Data

For this interface, it will display all about the transactions that are importing data from SAP MII and some essential details about this transactions for example : the execution date, the status and an important thing which is a description about the Error if the status is down cause this information will help the decision maker to have a general look about the problem and so take the best measures to find a solution. We can also download the log in any viewed interface by clicking the download button as the figure 4.16 shows.

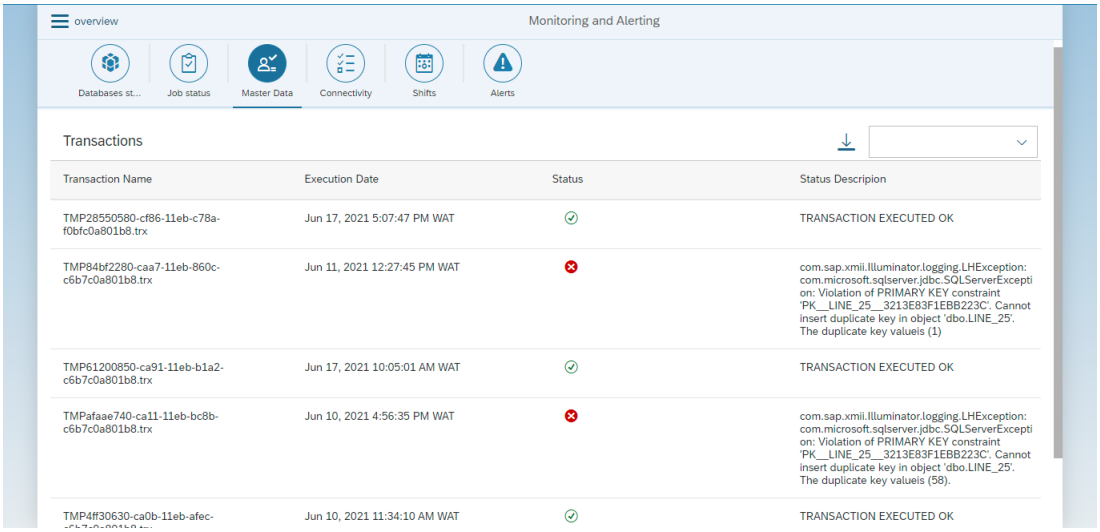


Figure 4.15: Master Data Interface

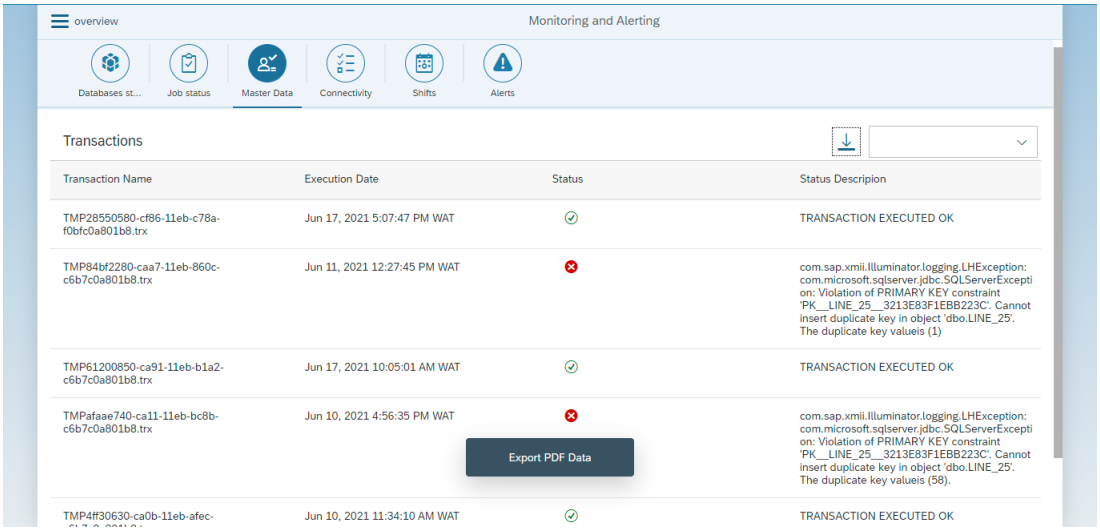


Figure 4.16: Master Data Interface : Downloading the Log

Connectivity

For the connectivity interface, it will be dealing with two main things, the first one as the figure 4.17 shows is PCo agents and the important things to know about them from their Status to a description about the error when needed, and the second one as shown in the figure 4.18 is the PCo transactions which are transactions that import data from PCo to the MII and for this transactions we will be displaying the execution time of every transaction over a period of time and also the status of this transaction. Another essential option that we had to include in this interface is the transaction details pop-up whee we can view a detailed error description about that transaction as the figure 4.19 shows.

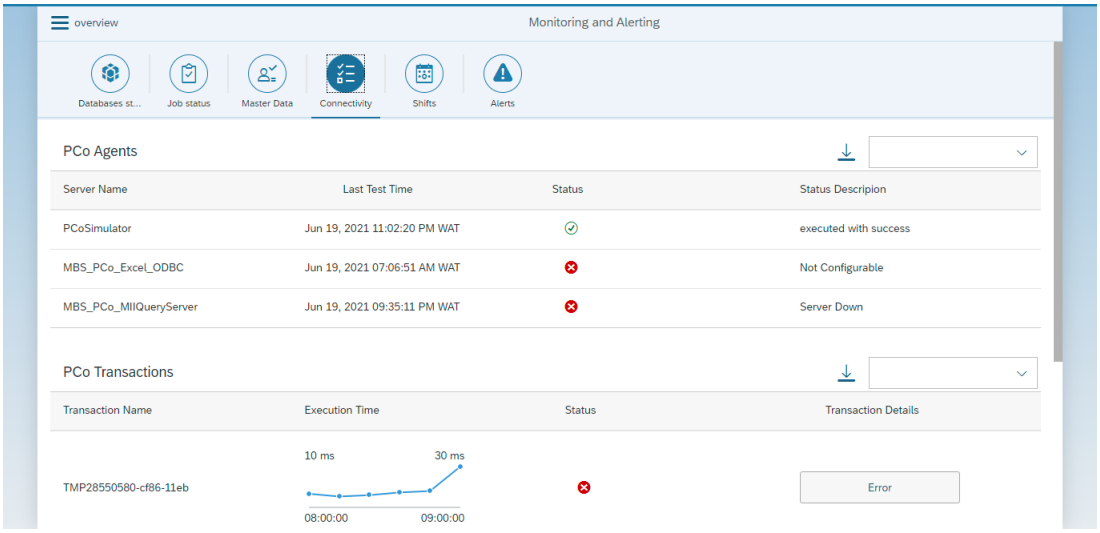


Figure 4.17: Connectivity Interface : PCo agents

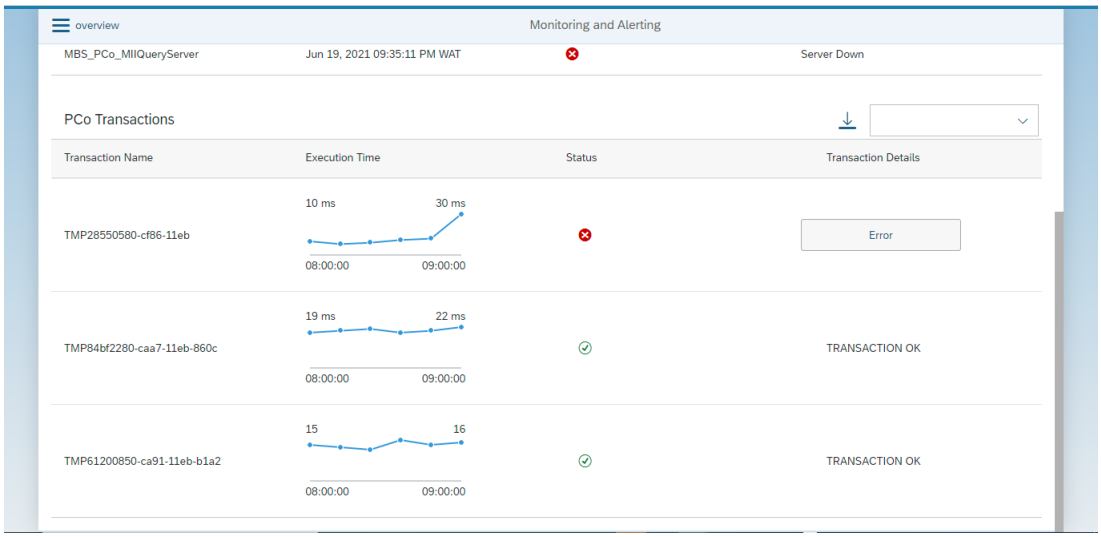


Figure 4.18: Connectivity Interface : PCo Transactions

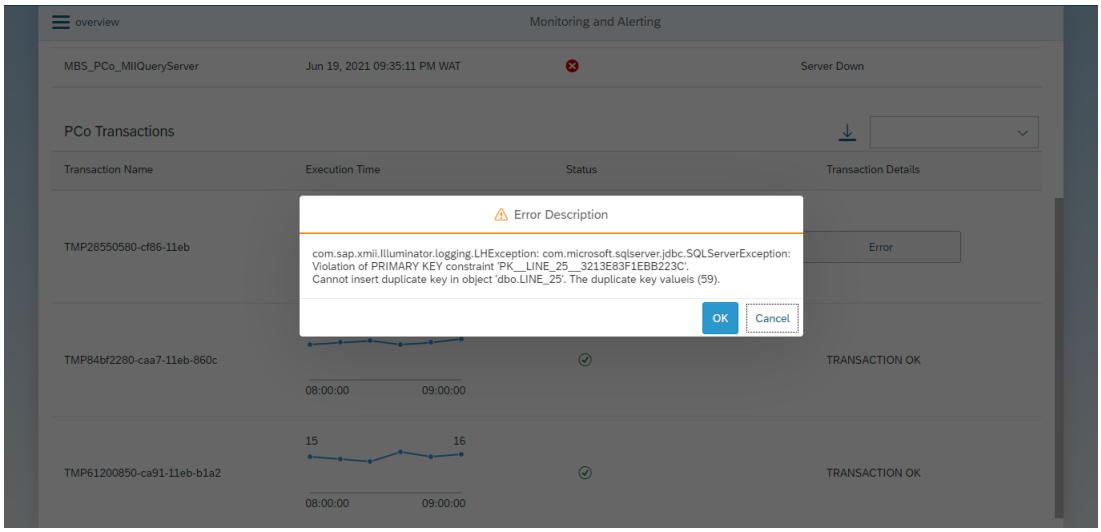


Figure 4.19: Connectivity Interface : Transaction Details

Shifts

As for this interface, a chart will resume the shifts done over a last period of time, and so for every day we will display the line which represents when the shift begins and when it ends and as a result we can make sure that all the shifts have been accomplished as scheduled and no time was lost let's say during the last week.

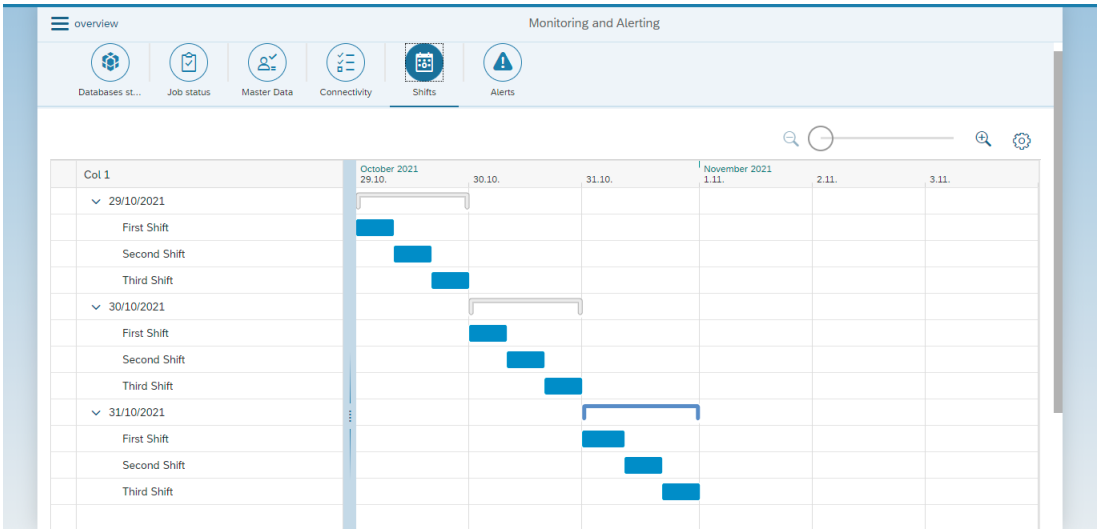


Figure 4.20: Shifts Interface

Alerts

For this last interface, it will display the errors accumulated in the other interfaces and so it will be like a pop-up messages to indicate that an error has occurred and something needs to be fixed. So the user when looking at this interface he can pay attention that something for example in the connectivity is having issues so he can navigate to that interface and examine the problem better. Also, if the the user can update the status of this alert for example if it is in progress or acknowledged or expired as shown in the drop box in the figure 4.22.

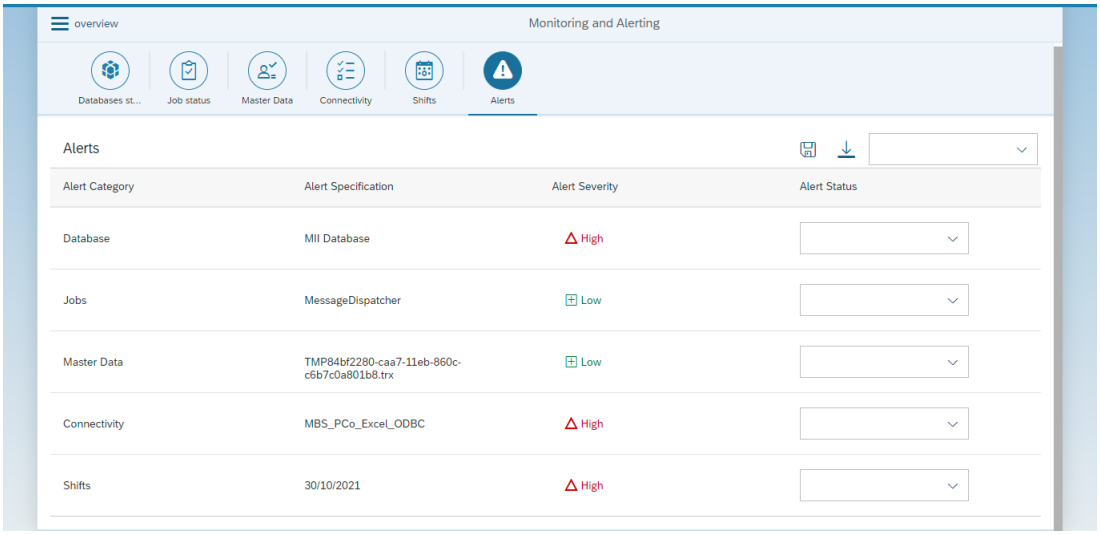


Figure 4.21: Alerts Interface

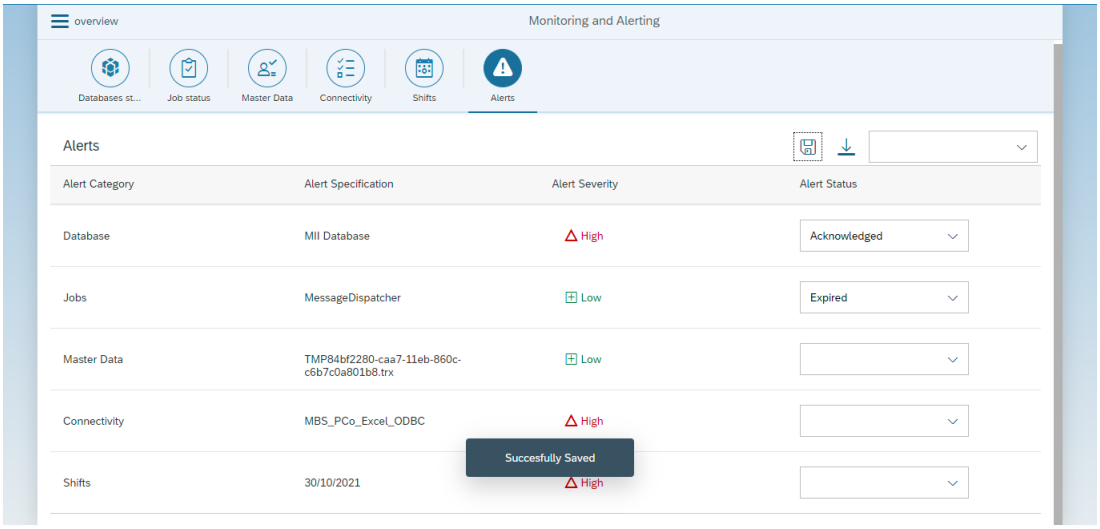


Figure 4.22: Alerts Interface : Updating the Status

All along this chapter, we dealt with the general environment of work at first, next we highlighted all the work that we have done from the importation of data either directly from the MII or from the PCo to the MII and then to the front-end part and we finished by displaying the interfaces developed during this work and all the functionalities that were put into those interfaces.

Chapter 5

Conclusion

As a general conclusion to all that we passed through in this report, i would like first to restate the idea of this project and most of all to make the solution established more comprehensible. So, the application developed all along this project will be with no doubt of use for any industrial company. In fact, any company would like to have an application that monitor all of the shop floor system and with shop floor we mean the plants and machines installed in the factory and also this application will monitor all the status that are essential to be aware of during production and another important thing that this application will be as a tool to be aware of any alert that is triggered and so the time of searching for the problem and locating it will be saved and as we always say: in industry, time means money.

However, in order to convey a more objective overview of this project, we have to state the limitations of this application that we found interesting to be aware of. Maybe the most important thing to mention here is that this application doesn't afford any steps to follow when dealing with an error so the decision making is all about the manager and the heads of the line of production in question. Another essential thing that we can't pass without highlighting it the fact that all the data is imported from the MII (manufacturing integration and intelligence) and the databases that it is linked to and as a result we have to make sure that those systems are always working perfectly so that the data projected into the application will be always updated.

So as for the further improvements that can added to this application, we can as we said earlier make like an intelligent decision making system that can give the user some sort of list of possible actions but we have to always leave the final decision to the human cause he has after all the better sight of his site and so he can have the better solution.

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